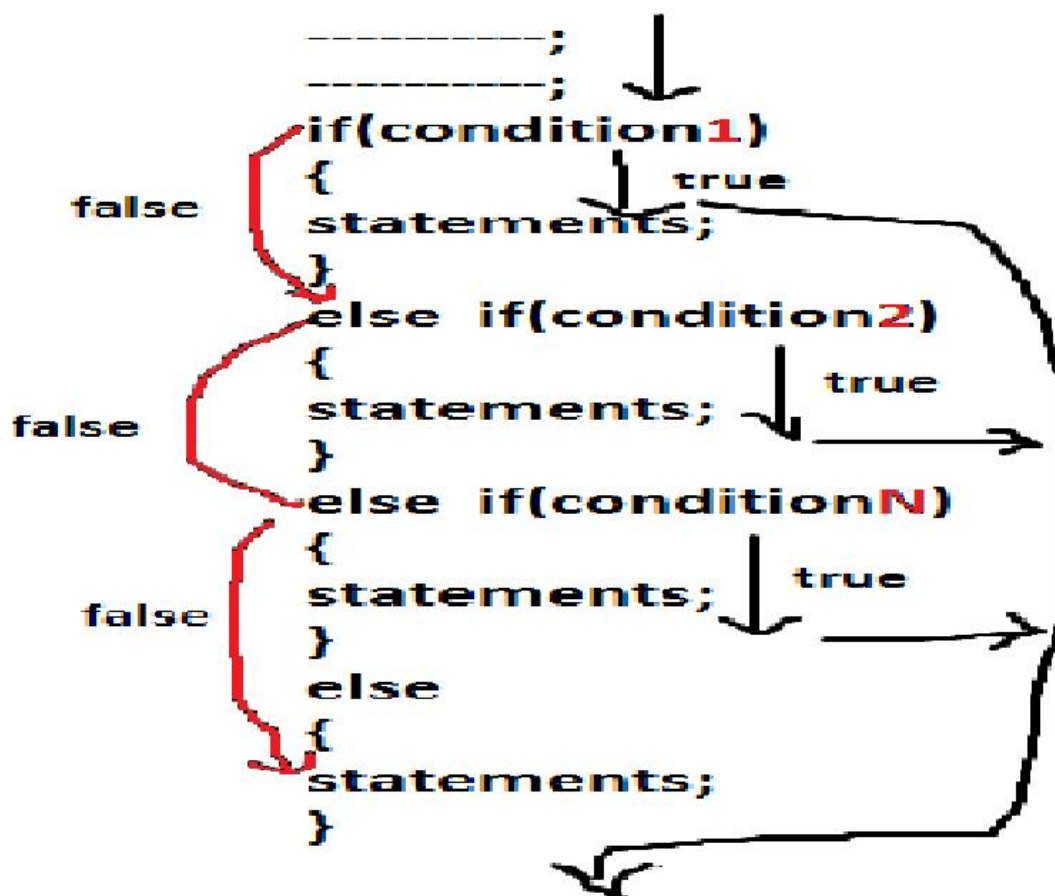
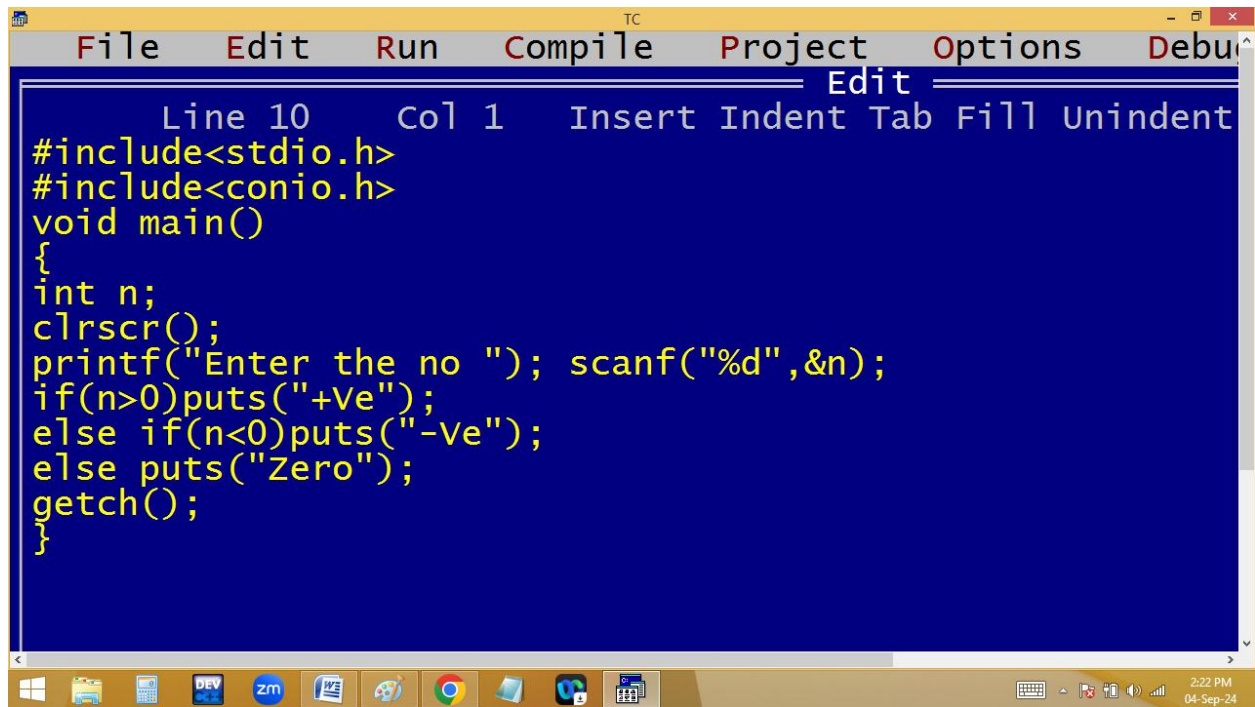


if...else if / ladder if

When the program contain 3 or more options and user have to select 1 then go for if else if ladder. In if..else if the conditions are tested from top to bottom one by one like the ladder steps and any one condition is true then statements in that if block are executed and program execution jumped to outside the if series. If no condition true then statements in else are executed. Here else is optional.

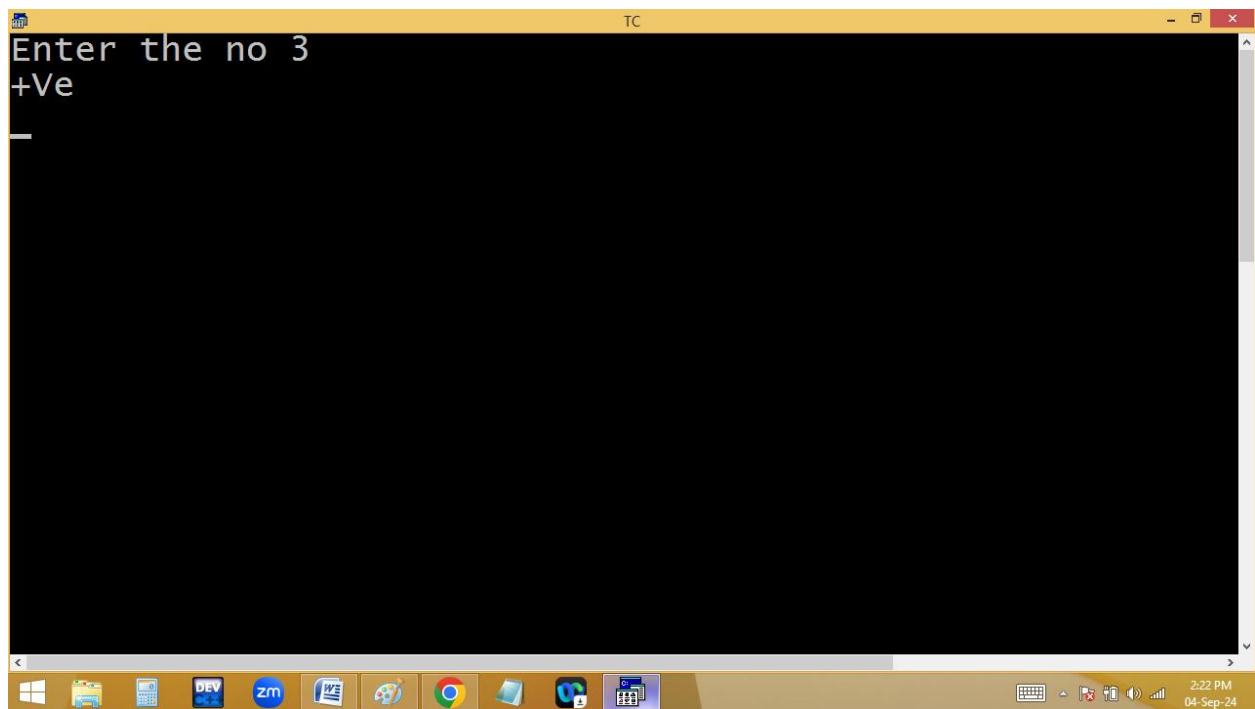


Finding +ve/-ve/0:



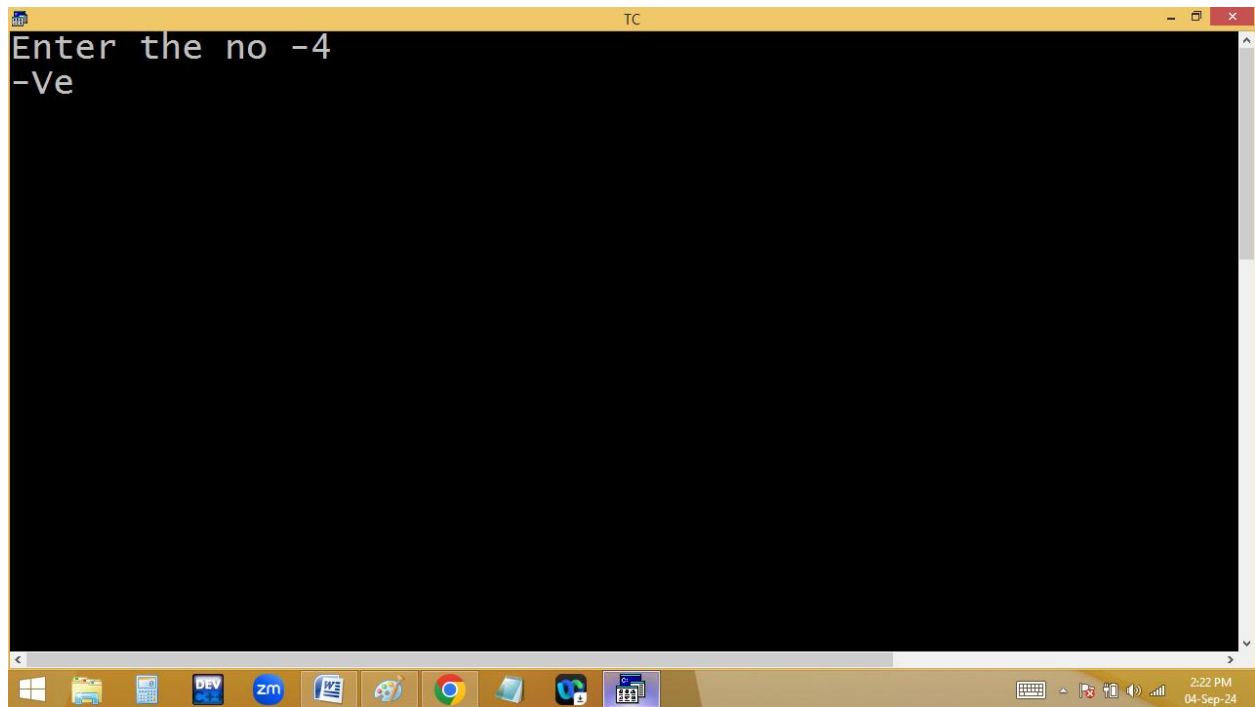
The screenshot shows the Turbo C++ (TC) editor window. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates 'Line 10 Col 1' and lists editing options: Insert, Indent, Tab, Fill, and Unindent. The code is written in yellow text on a blue background. It includes the standard input/output and console I/O headers, defines the main function, declares an integer variable 'n', clears the screen, prompts the user for a number, reads the input, and uses conditional statements to print '+Ve', '-Ve', or 'Zero' based on the sign of 'n'. The program ends with a getch() call to pause execution.

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int n;
    clrscr();
    printf("Enter the no "); scanf("%d",&n);
    if(n>0)puts("+Ve");
    else if(n<0)puts("-Ve");
    else puts("Zero");
    getch();
}
```



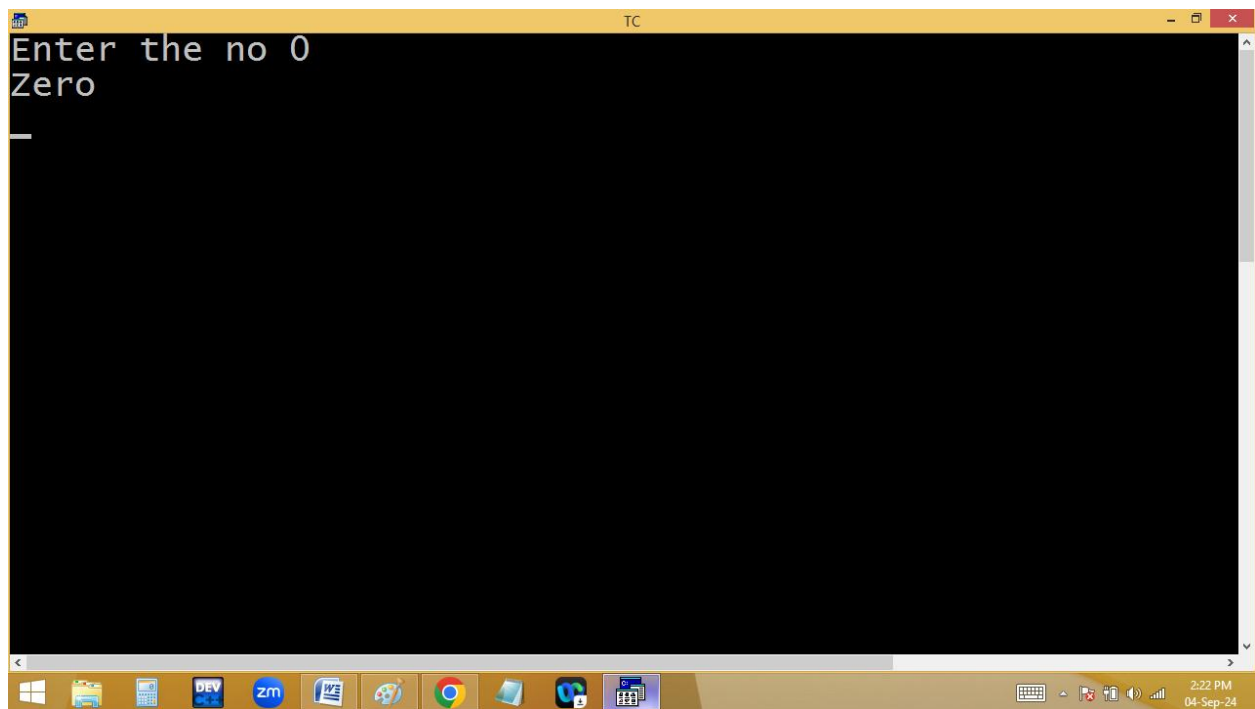
The screenshot shows the Turbo C++ (TC) editor window after the program has been executed. The output is displayed in white text on a black background. The prompt 'Enter the no' is followed by the user input '3'. The program has printed '+Ve' on the next line, indicating that the input number is positive. The status bar at the bottom right shows the time as 2:22 PM on 04-Sep-24.

```
Enter the no 3
+Ve
```



```
TC
Enter the no -4
-Ve
```

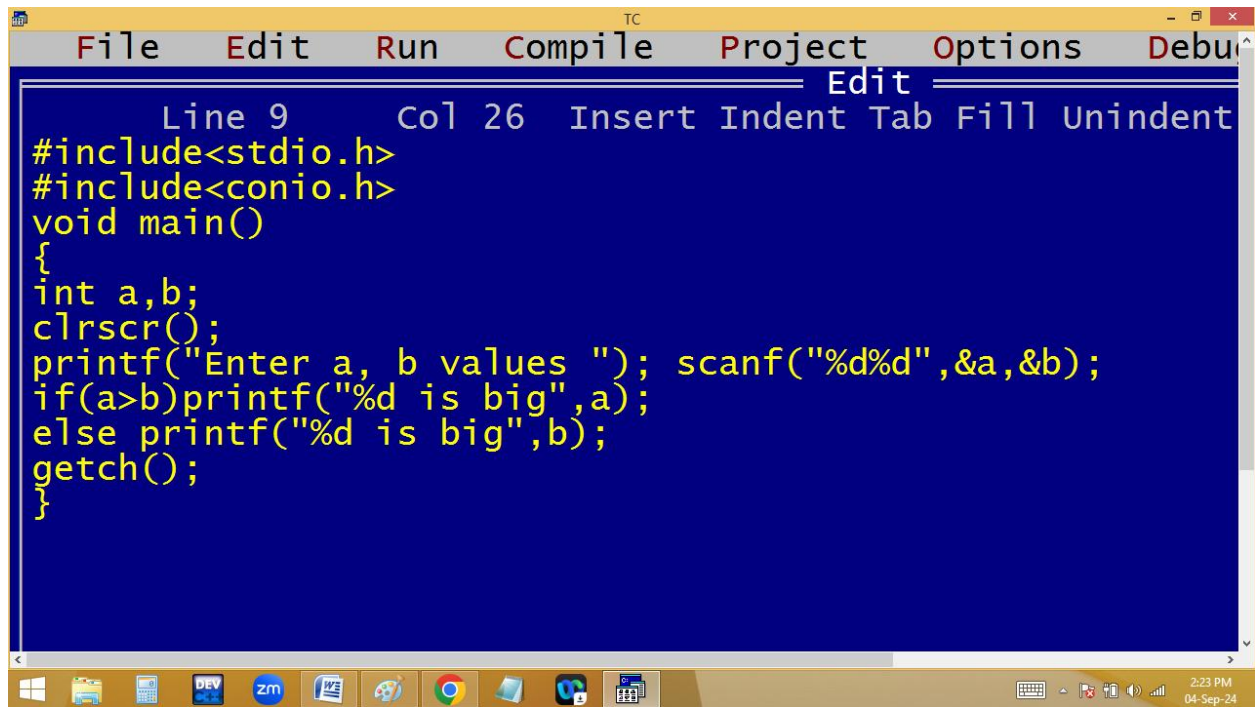
This screenshot shows a Windows terminal window with a yellow title bar labeled 'TC'. The terminal has a black background with white text. The prompt 'Enter the no -4' is displayed on the first line, and the input '-Ve' is on the second line. The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 2:22 PM on 04-Sep-24.



```
TC
Enter the no 0
Zero
```

This screenshot shows a similar Windows terminal window titled 'TC'. The prompt 'Enter the no 0' is displayed on the first line, and the input 'Zero' is on the second line. The terminal window and taskbar are identical to the one above.

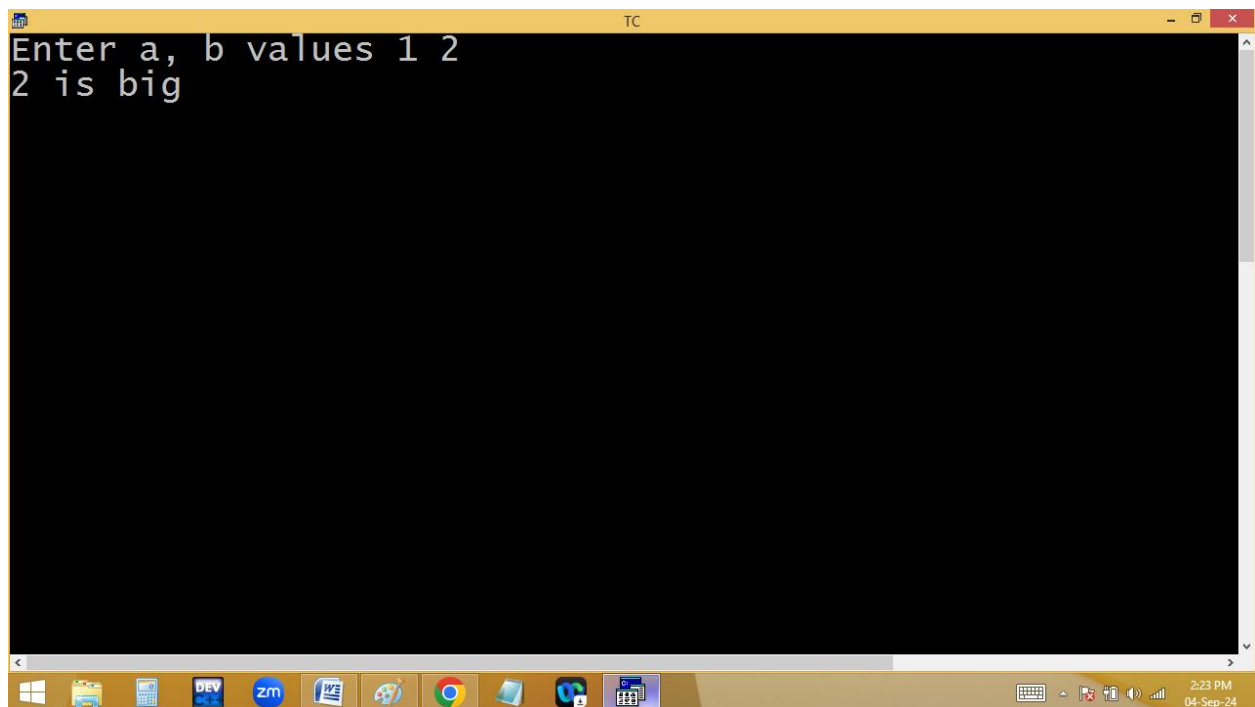
Finding max in 2 no's:



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug) and a toolbar. The main editing area has a blue background and contains the following C code:

```
Line 9      Col 26  Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b;
clrscr();
printf("Enter a, b values "); scanf("%d%d",&a,&b);
if(a>b)printf("%d is big",a);
else printf("%d is big",b);
getch();
}
```

The Windows taskbar at the bottom shows various application icons and the system clock indicating 2:23 PM on 04-Sep-24.

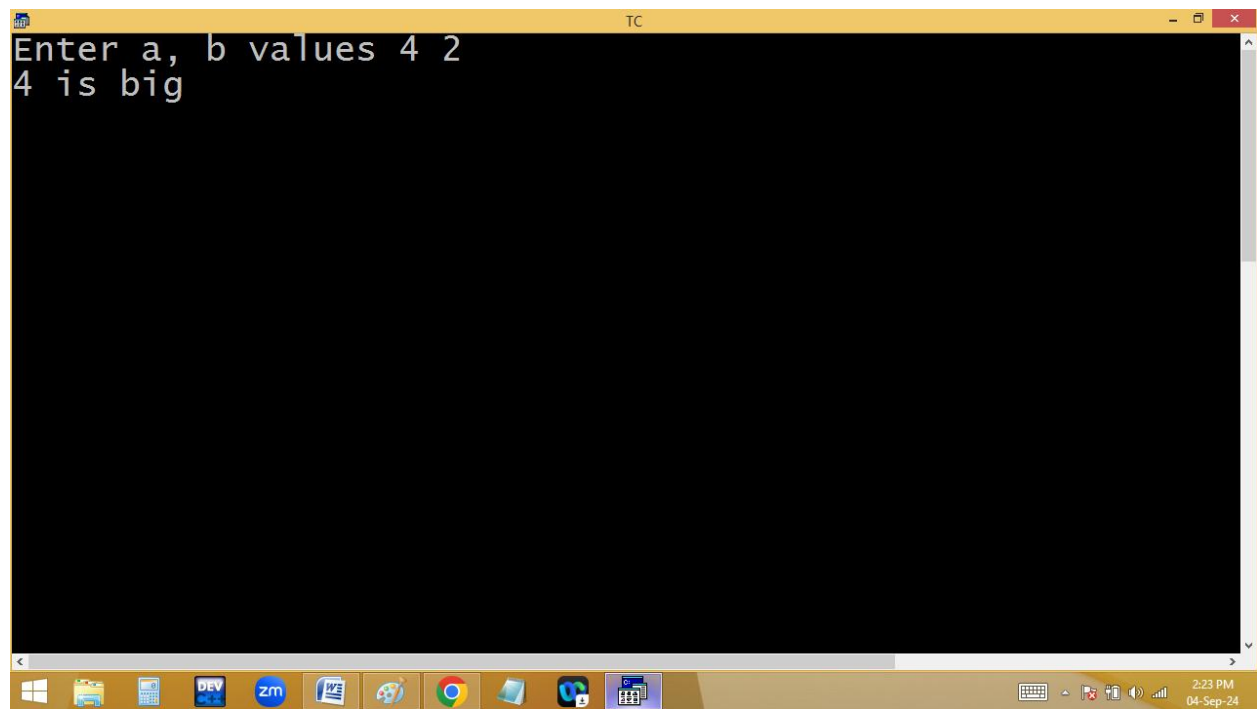


The screenshot shows the Turbo C++ (TC) IDE with a black background, displaying the output of the program. The text "Enter a, b values 1 2" is on the first line, and "2 is big" is on the second line, indicating that the program correctly identified 2 as the maximum value.

```
Enter a, b values 1 2
2 is big
```

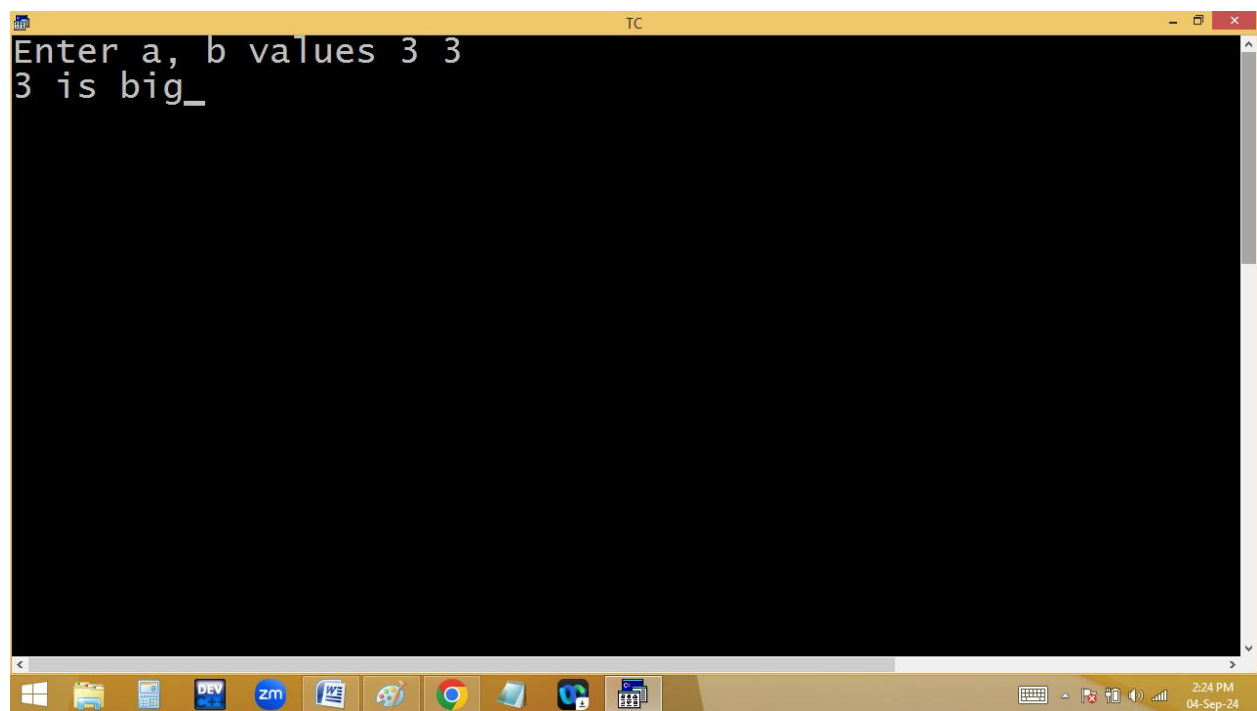
The Windows taskbar at the bottom is identical to the first screenshot, showing the same application icons and system clock (2:23 PM on 04-Sep-24).

```
TC
Enter a, b values 4 2
4 is big
```

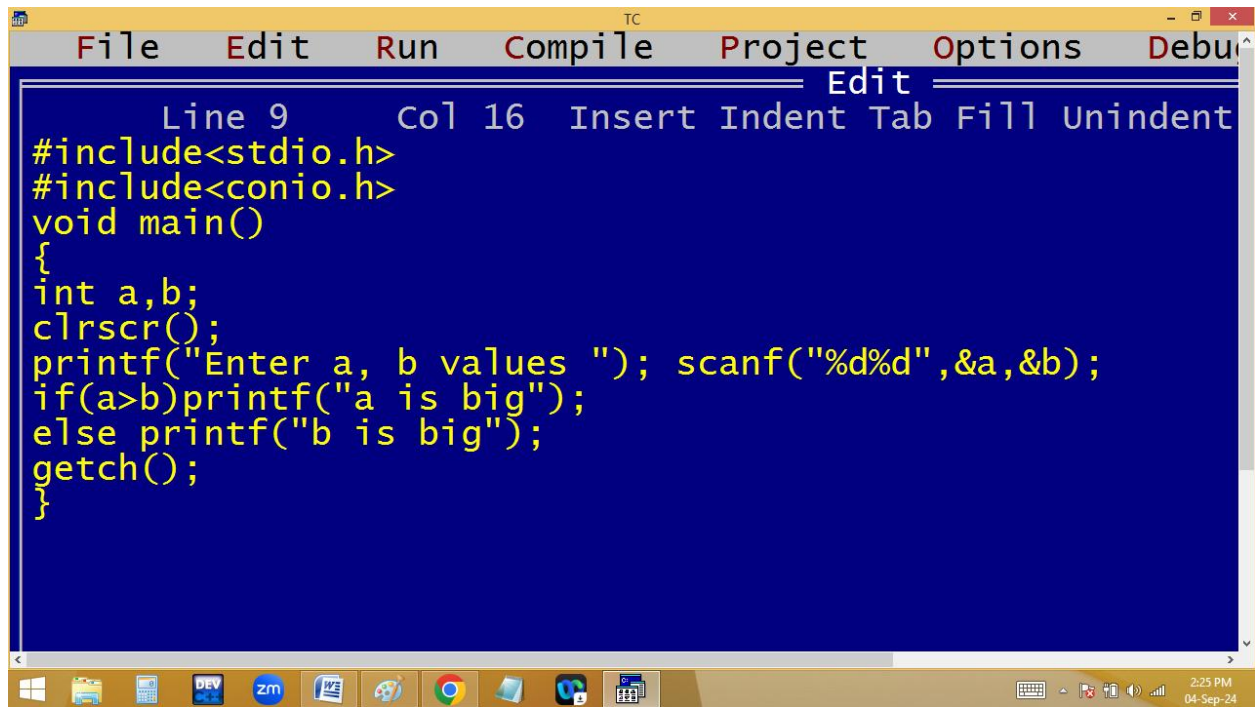


The screenshot shows a Turbo C++ (TC) IDE window. The title bar is yellow and contains the text 'TC'. The main window is black with white text. The text 'Enter a, b values 4 2' is on the first line, and '4 is big' is on the second line. The Windows taskbar is visible at the bottom, showing various icons including Windows, File Explorer, DEV, zm, Word, Paint, Chrome, and a calendar icon. The system clock in the bottom right corner shows '2:23 PM' and '04-Sep-24'.

```
TC
Enter a, b values 3 3
3 is big_
```



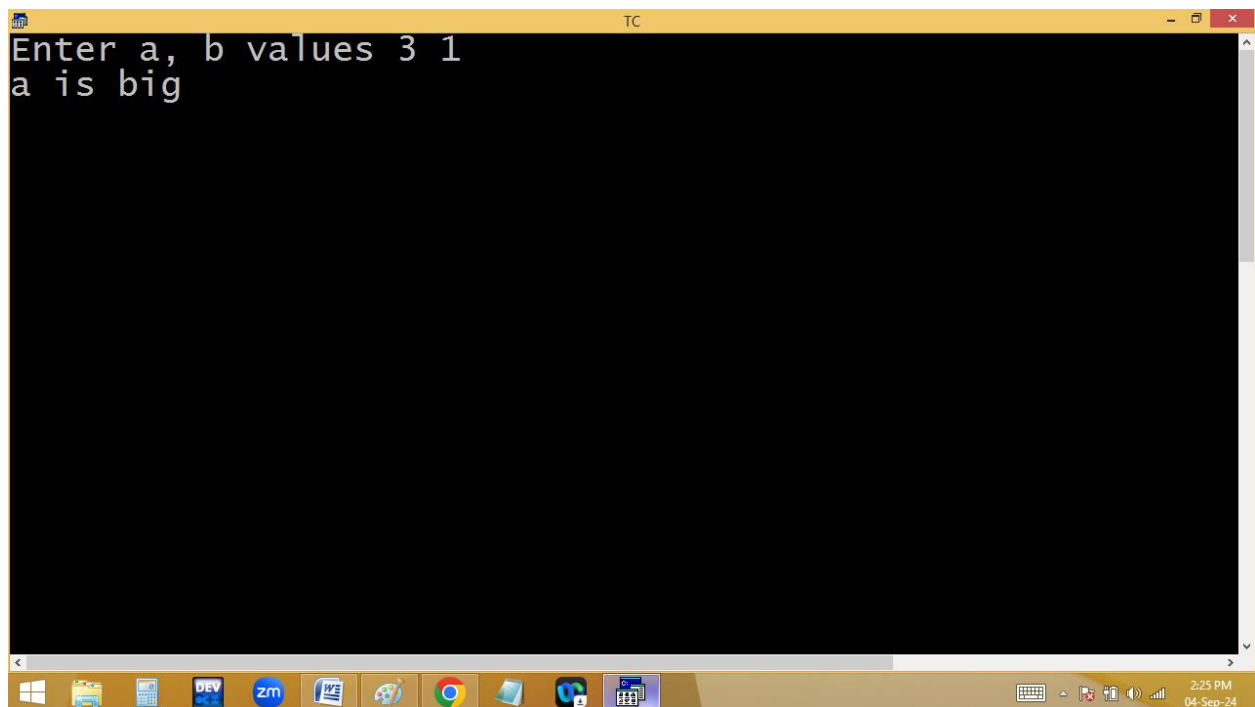
The screenshot shows a Turbo C++ (TC) IDE window. The title bar is yellow and contains the text 'TC'. The main window is black with white text. The text 'Enter a, b values 3 3' is on the first line, and '3 is big_' is on the second line. The Windows taskbar is visible at the bottom, showing various icons including Windows, File Explorer, DEV, zm, Word, Paint, Chrome, and a calendar icon. The system clock in the bottom right corner shows '2:24 PM' and '04-Sep-24'.



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug) and a toolbar. The main editing area has a blue background and contains the following C code:

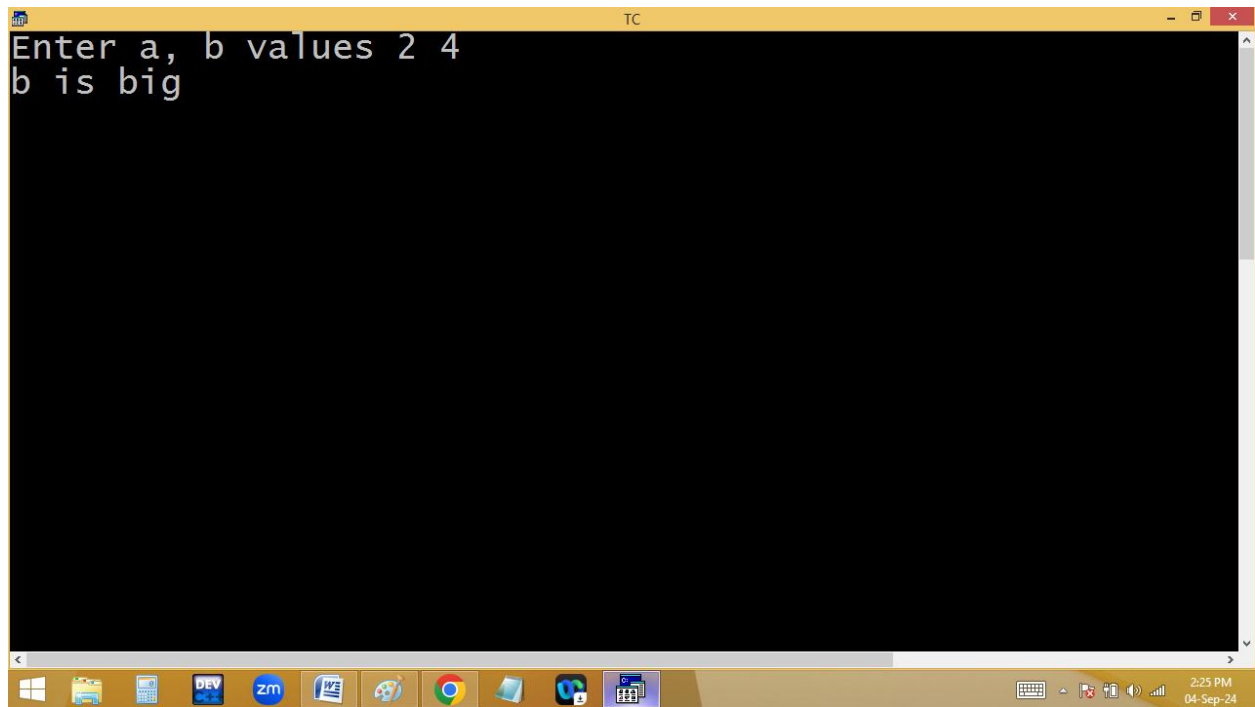
```
Line 9      Col 16  Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
    int a,b;
    clrscr();
    printf("Enter a, b values "); scanf("%d%d",&a,&b);
    if(a>b)printf("a is big");
    else printf("b is big");
    getch();
}
```

The Windows taskbar at the bottom shows various application icons and the system clock indicating 2:25 PM on 04-Sep-24.



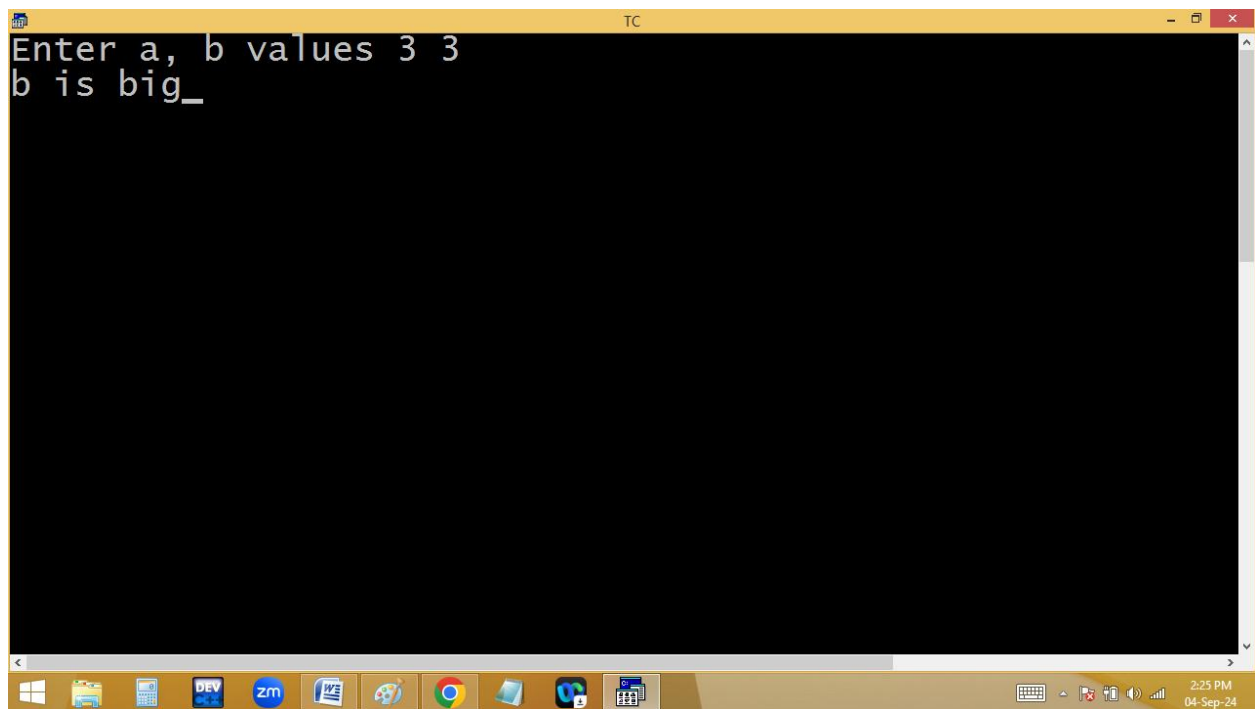
The screenshot shows the Turbo C++ (TC) IDE with a black background, displaying the output of the program. The text "Enter a, b values 3 1" and "a is big" is visible in white. The Windows taskbar at the bottom is identical to the first screenshot, showing the same application icons and system clock (2:25 PM on 04-Sep-24).

```
TC
Enter a, b values 2 4
b is big
```

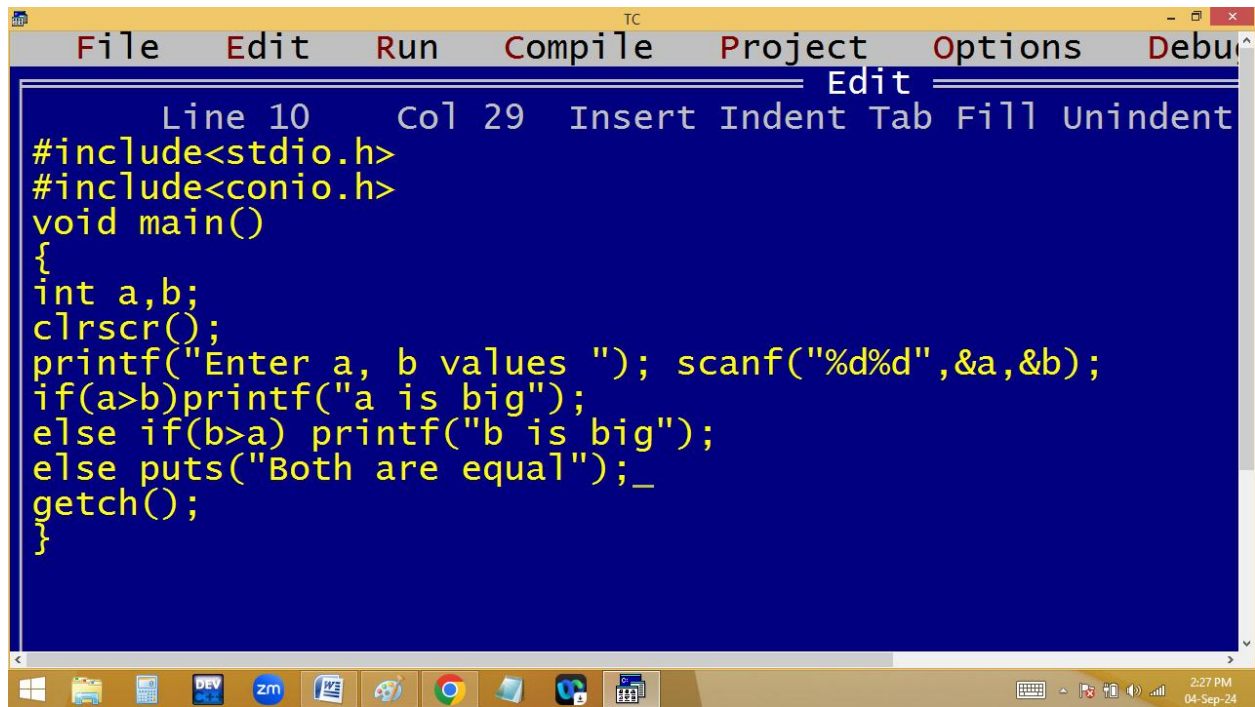


The screenshot shows a Turbo C++ IDE window titled 'TC'. The main text area contains the following code:
`Enter a, b values 2 4`
`b is big`
The window has a yellow title bar and a black text area. The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 2:25 PM on 04-Sep-24.

```
TC
Enter a, b values 3 3
b is big_
```



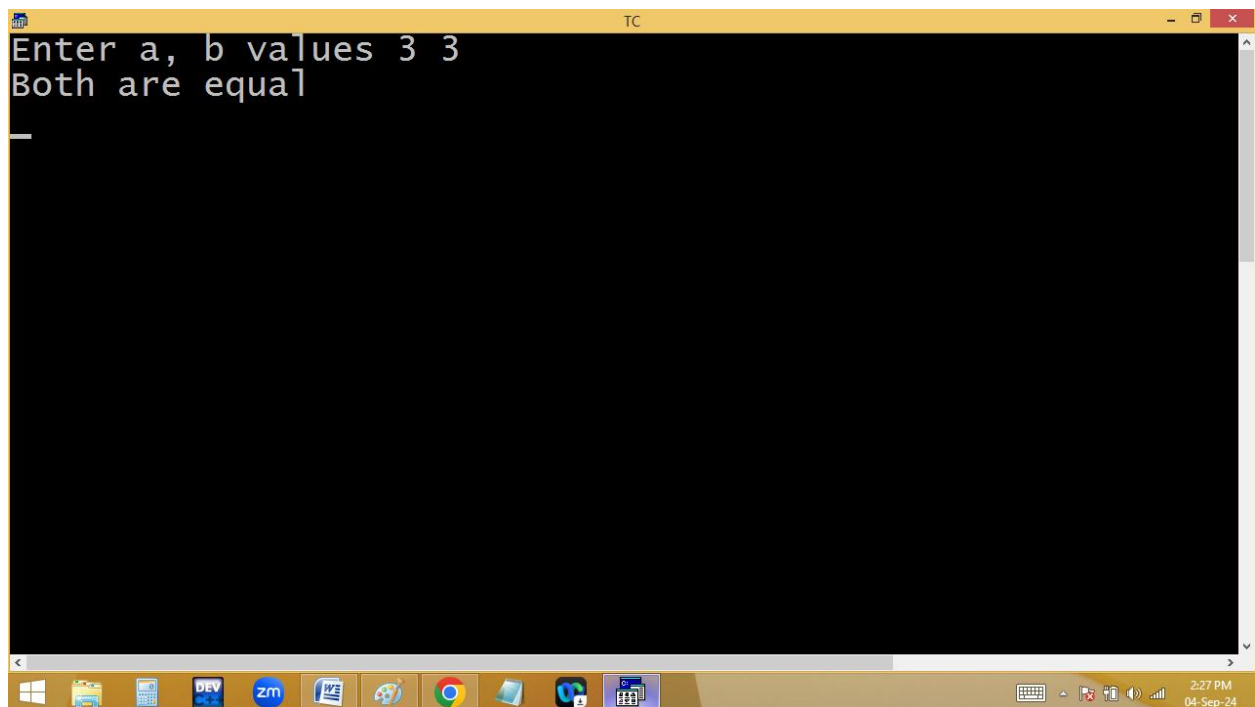
The screenshot shows a Turbo C++ IDE window titled 'TC'. The main text area contains the following code:
`Enter a, b values 3 3`
`b is big_`
The window has a yellow title bar and a black text area. The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 2:25 PM on 04-Sep-24.



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug) and a toolbar. The main window displays a C program for comparing two numbers. The code is as follows:

```
Line 10    Col 29  Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b;
clrscr();
printf("Enter a, b values "); scanf("%d%d",&a,&b);
if(a>b)printf("a is big");
else if(b>a) printf("b is big");
else puts("Both are equal");_
getch();
}
```

The Windows taskbar at the bottom shows various application icons and the system clock indicating 2:27 PM on 04-Sep-24.

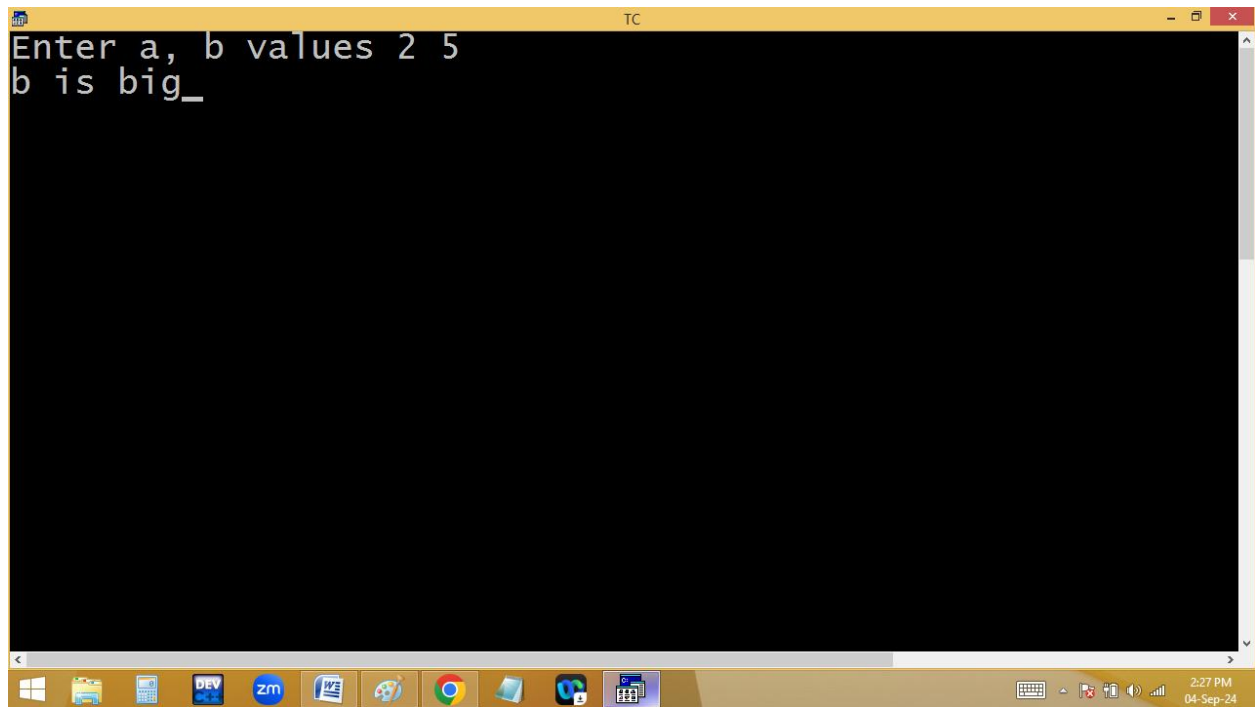


The screenshot shows the Turbo C++ (TC) IDE with the same menu bar and toolbar. The main window displays the output of the program. The text shown is:

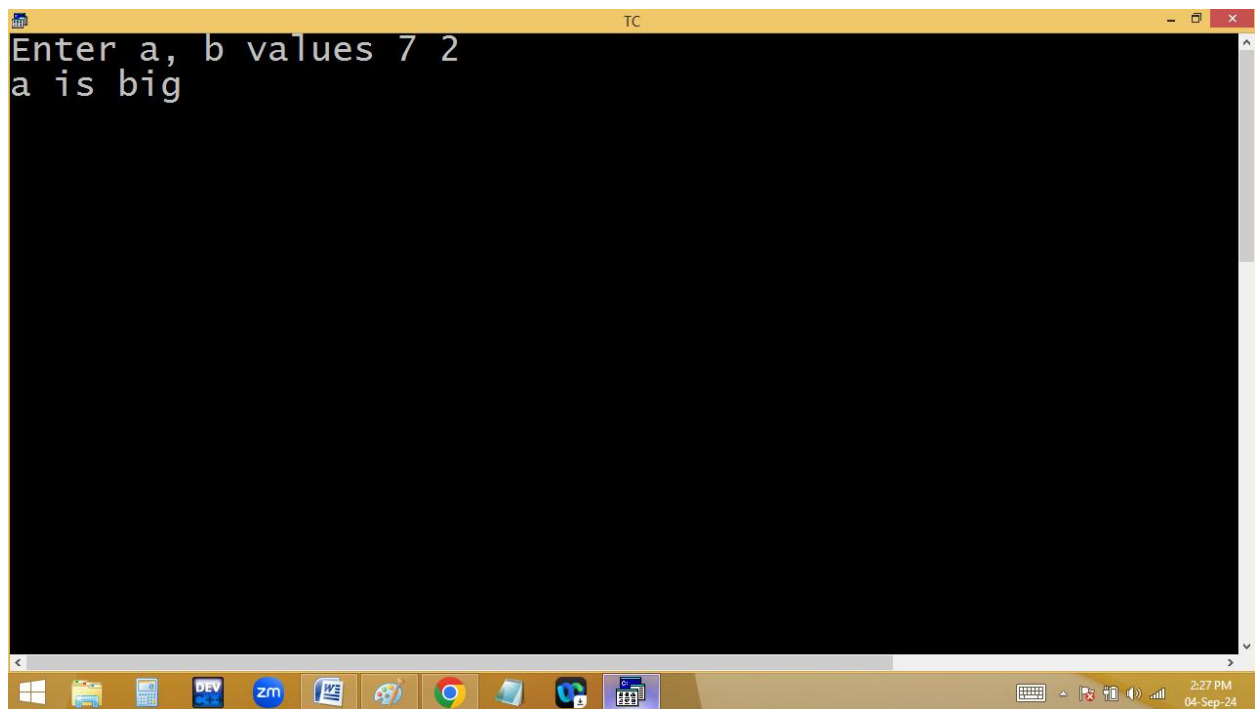
```
Enter a, b values 3 3
Both are equal
```

The Windows taskbar at the bottom shows various application icons and the system clock indicating 2:27 PM on 04-Sep-24.

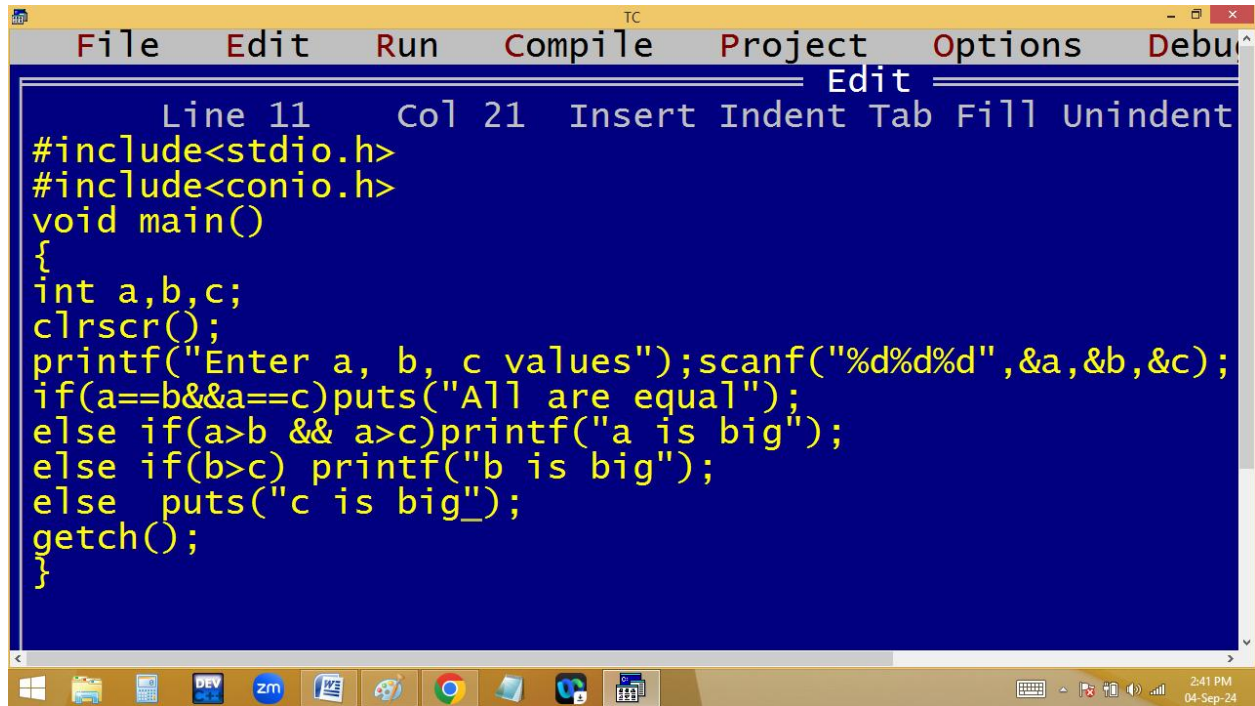

```
TC
Enter a, b values 2 5
b is big_
```



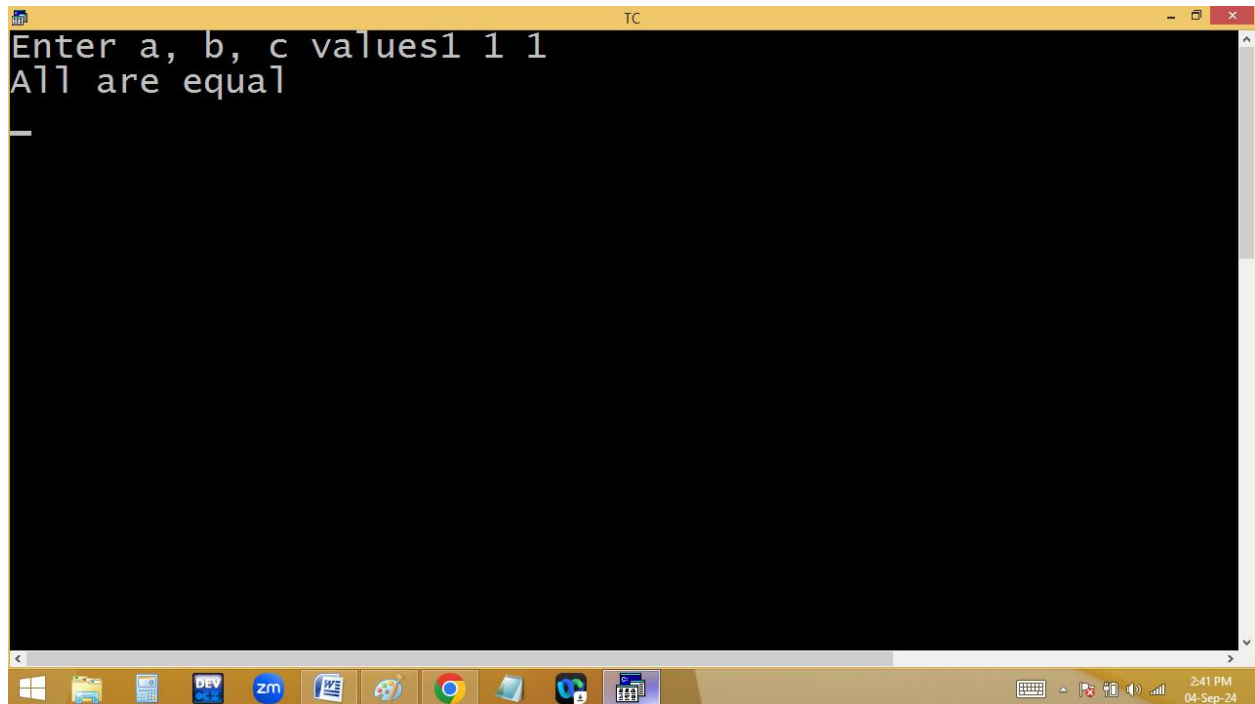
```
TC
Enter a, b values 7 2
a is big
```



Finding max in 3 no's:



```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 21 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c;
clrscr();
printf("Enter a, b, c values");scanf("%d%d%d",&a,&b,&c);
if(a==b&&a==c)puts("All are equal");
else if(a>b && a>c)printf("a is big");
else if(b>c) printf("b is big");
else puts("c is big");
getch();
}
```

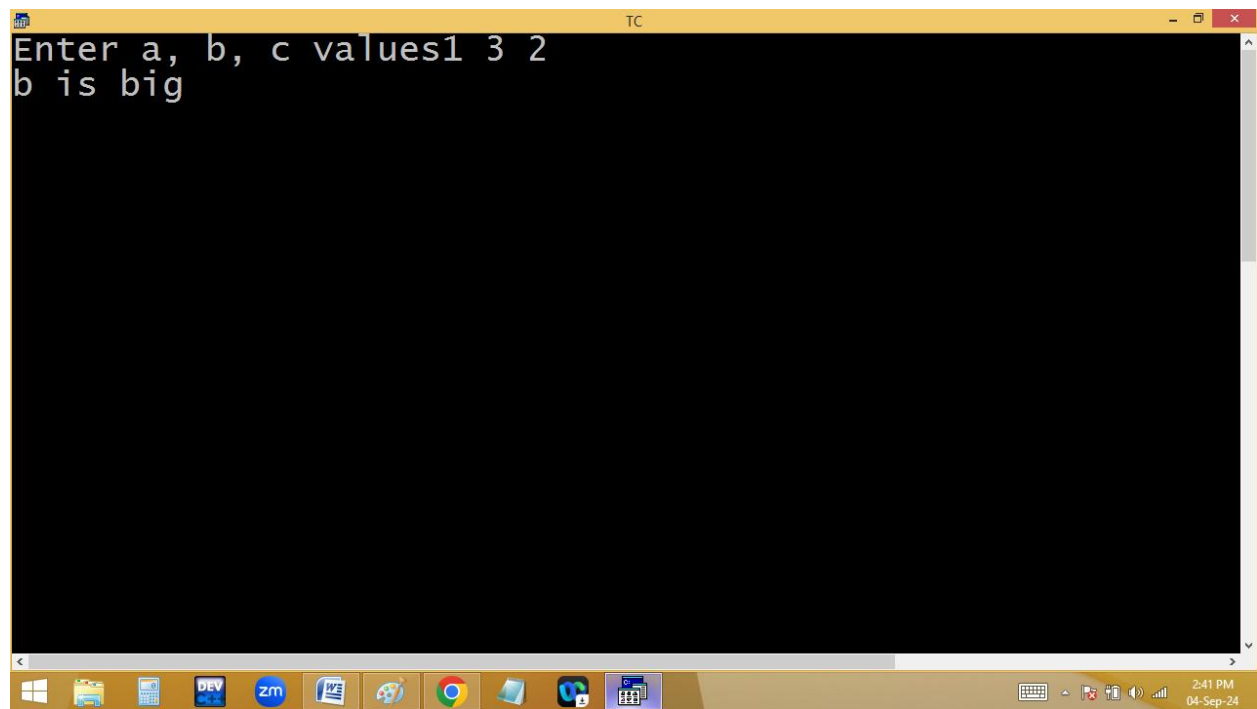


```
TC
Enter a, b, c values1 1 1
All are equal
```

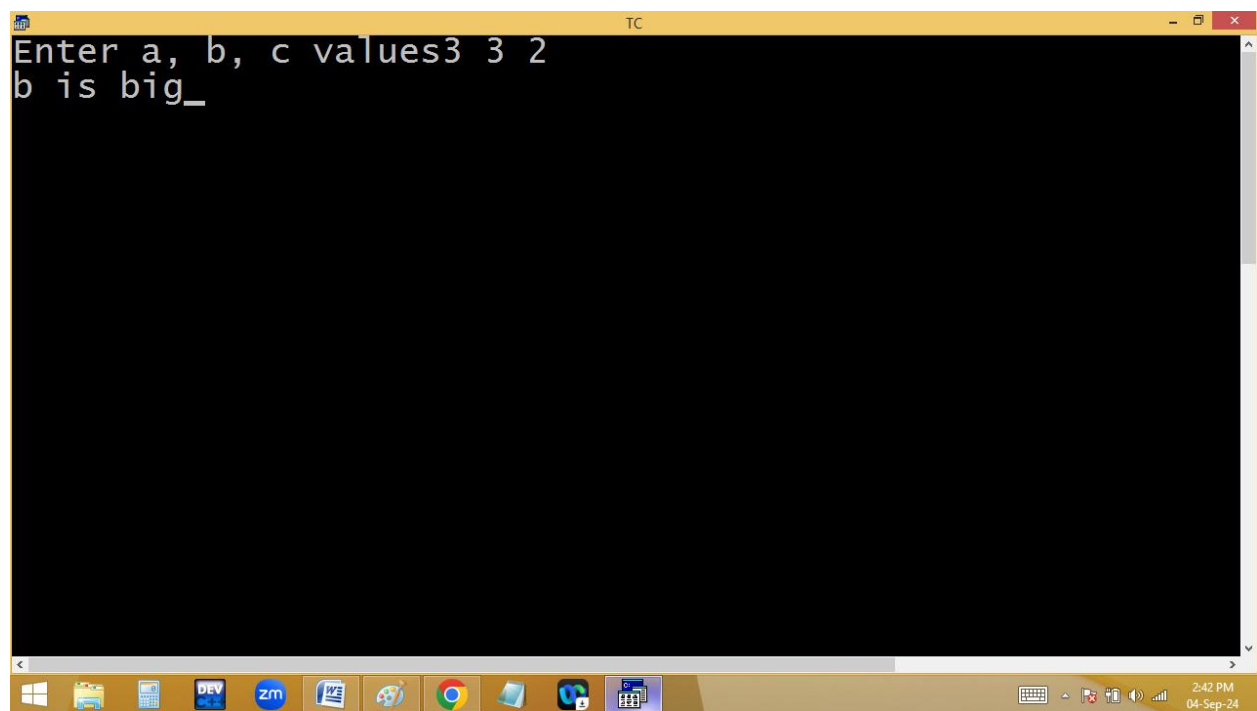
```
TC
Enter a, b, c values1 2 3
c is big
```

```
TC
Enter a, b, c values3 2 1
a is big
```

```
TC
Enter a, b, c values1 3 2
b is big
```



```
TC
Enter a, b, c values3 3 2
b is big_
```



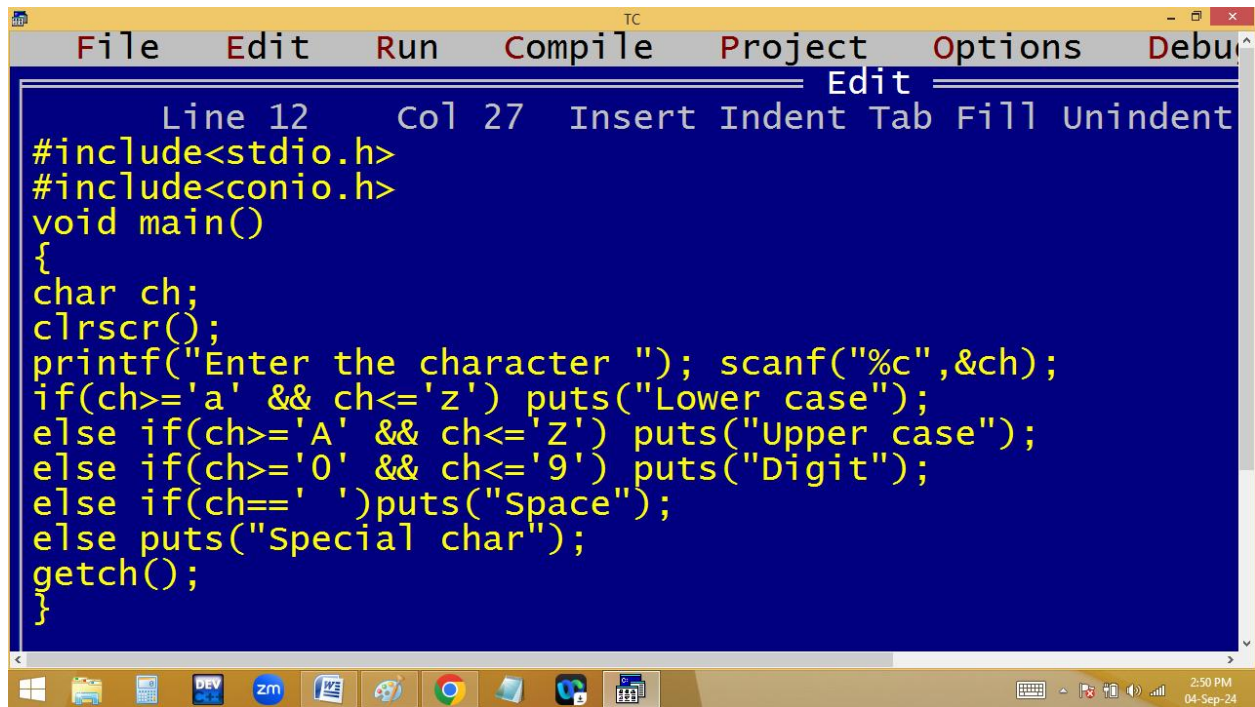
```
TC
Line 17 Col 45 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c;
clrscr();
printf("Enter a, b, c values");scanf("%d%d%d",&a,&b,&c);
if(a==b&&a==c)puts("All are equal");
else if(a>b && a>c)printf("a is big");
else if(a==b && b>c) printf("a & b are big");
else if(b==c && b>a) printf("b & c are big");
else if(a==c && a>b)printf("a & c are big");
else if(b>c) printf("b is big");
else puts("c is big");
getch();
}
```

```
TC
Enter a, b, c values2 1 2
a & c are big_
```

```
TC
Enter a, b, c values4 4 4
All are equal
```

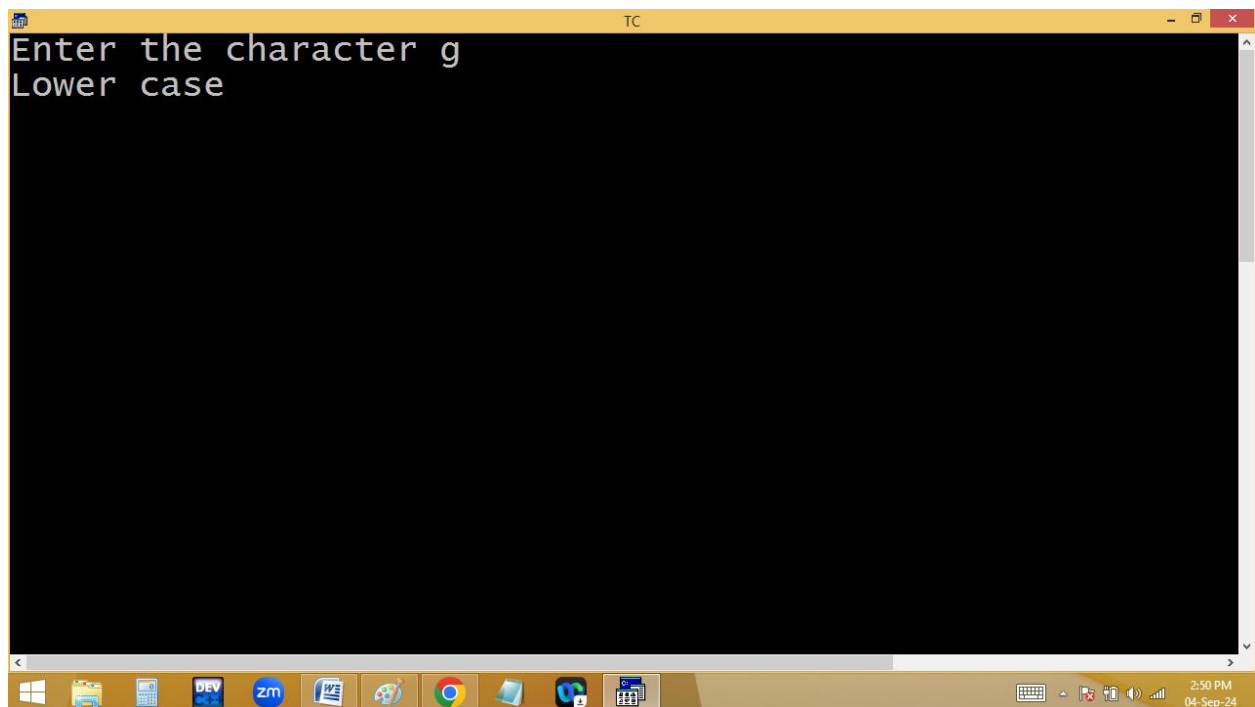
```
TC
Enter a, b, c values1 2 3
c is big
```

Finding lower / upper / digit / space / special char?



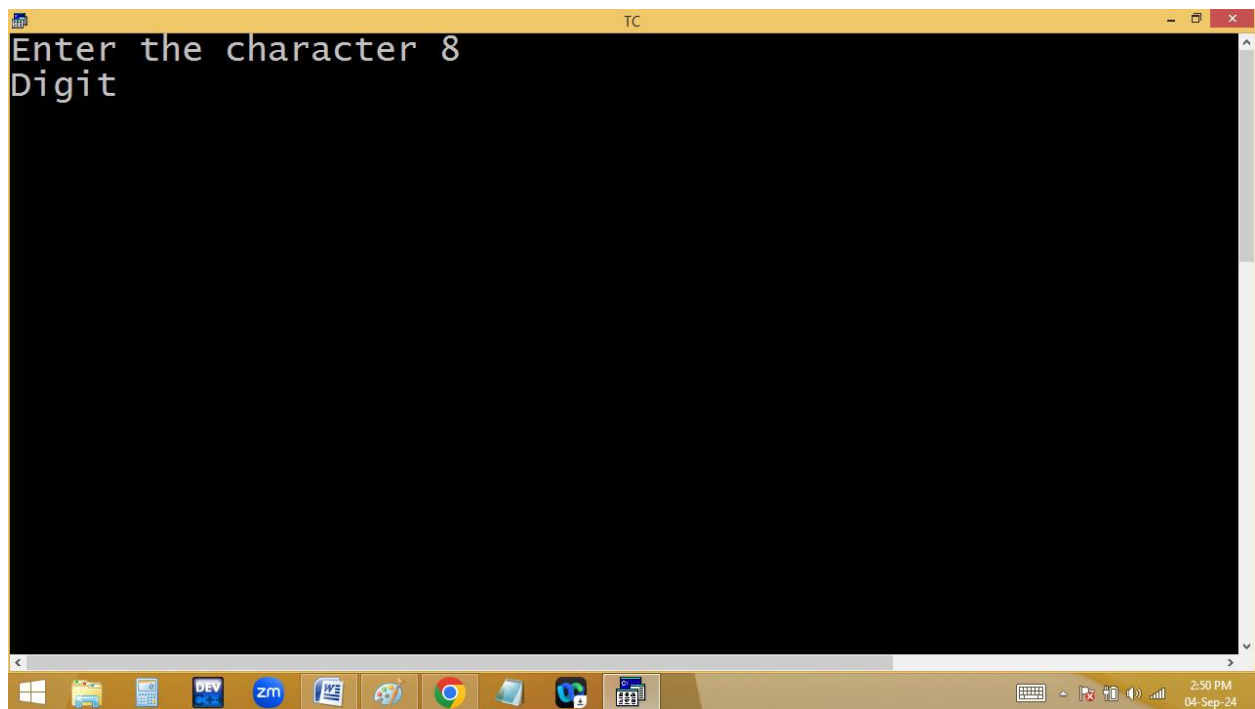
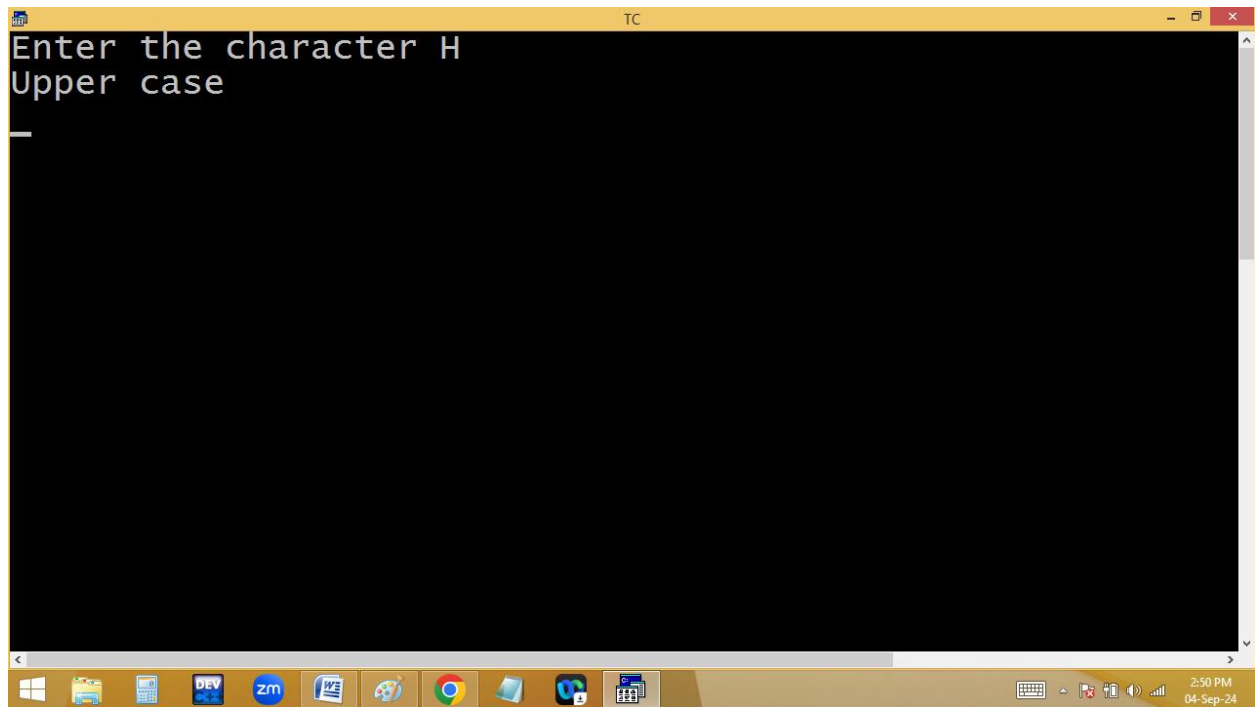
The screenshot shows the Turbo C++ (TC) editor window. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates 'Line 12 Col 27' and provides options like Insert, Indent, Tab, Fill, and Unindent. The code is written in yellow text on a blue background. The program includes `<stdio.h>` and `<conio.h>`, defines a `main` function, declares a `char ch`, clears the screen with `clrscr()`, and prompts the user to 'Enter the character'. It then uses a series of `if` statements to check if the character is lowercase, uppercase, a digit, a space, or a special character, and prints the corresponding message. Finally, it uses `getch()` to wait for a key press before exiting.

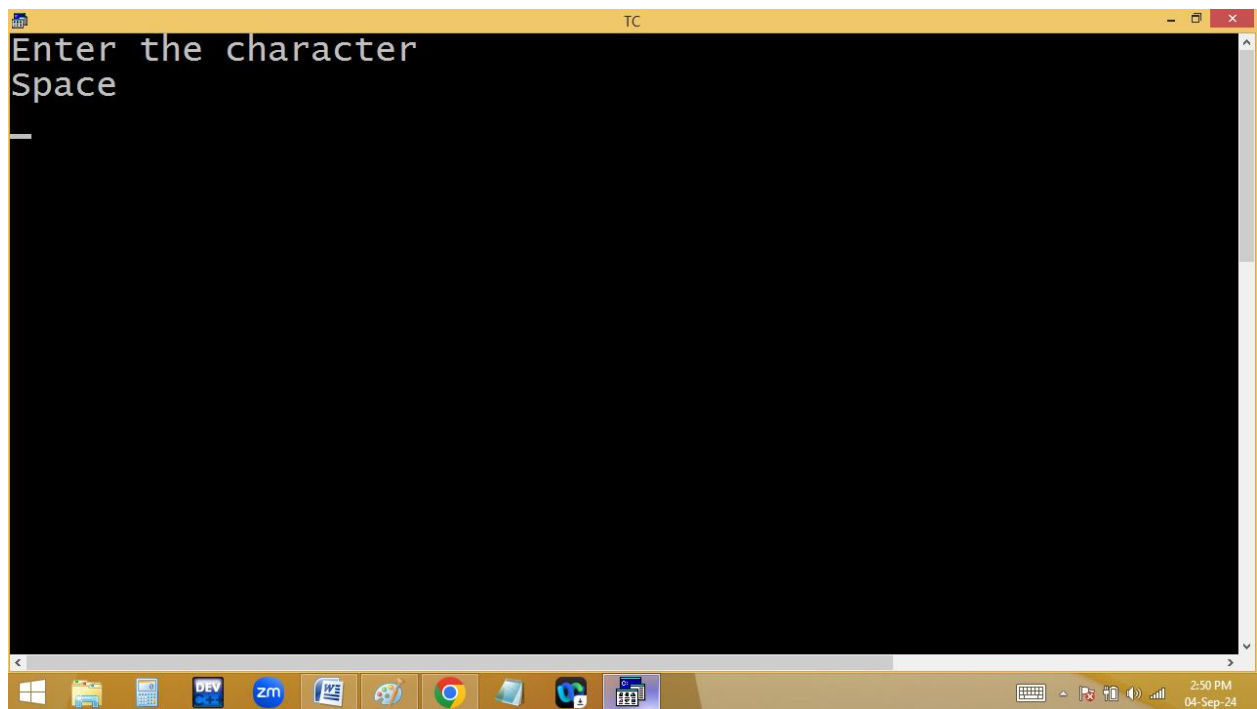
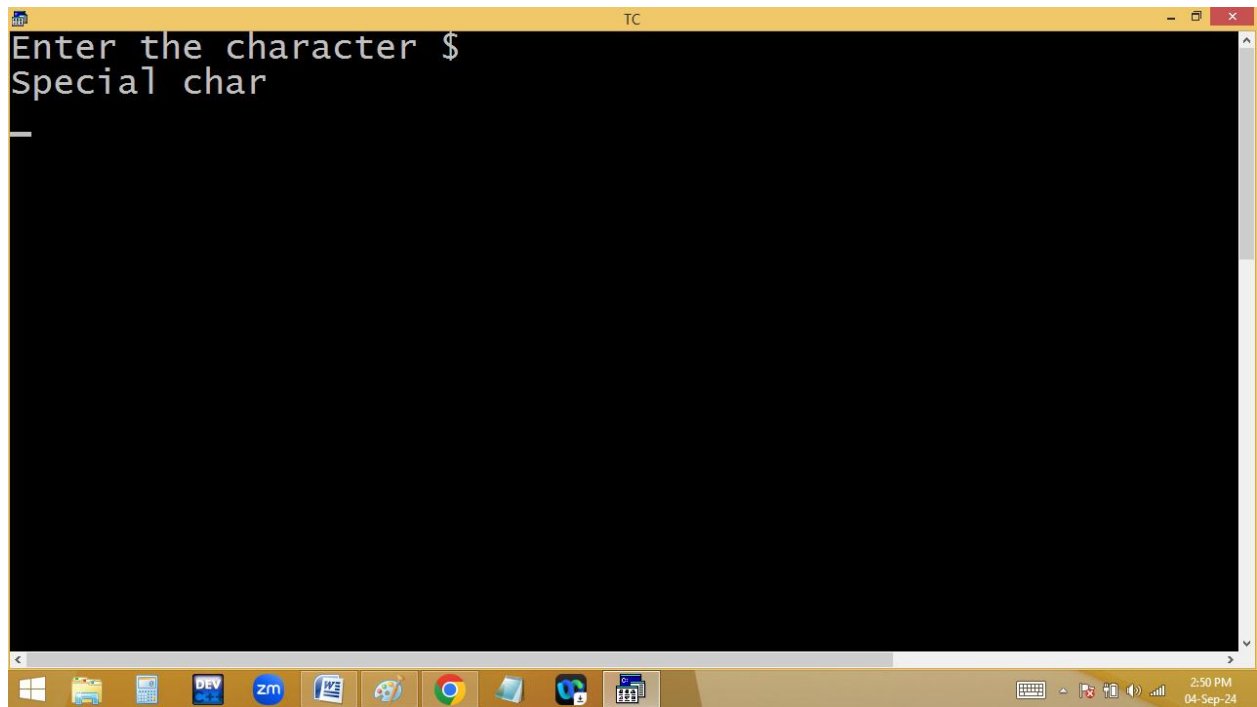
```
Line 12 Col 27 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); scanf("%c",&ch);
if(ch>='a' && ch<='z') puts("Lower case");
else if(ch>='A' && ch<='Z') puts("Upper case");
else if(ch>='0' && ch<='9') puts("Digit");
else if(ch==' ')puts("Space");
else puts("Special char");
getch();
}
```

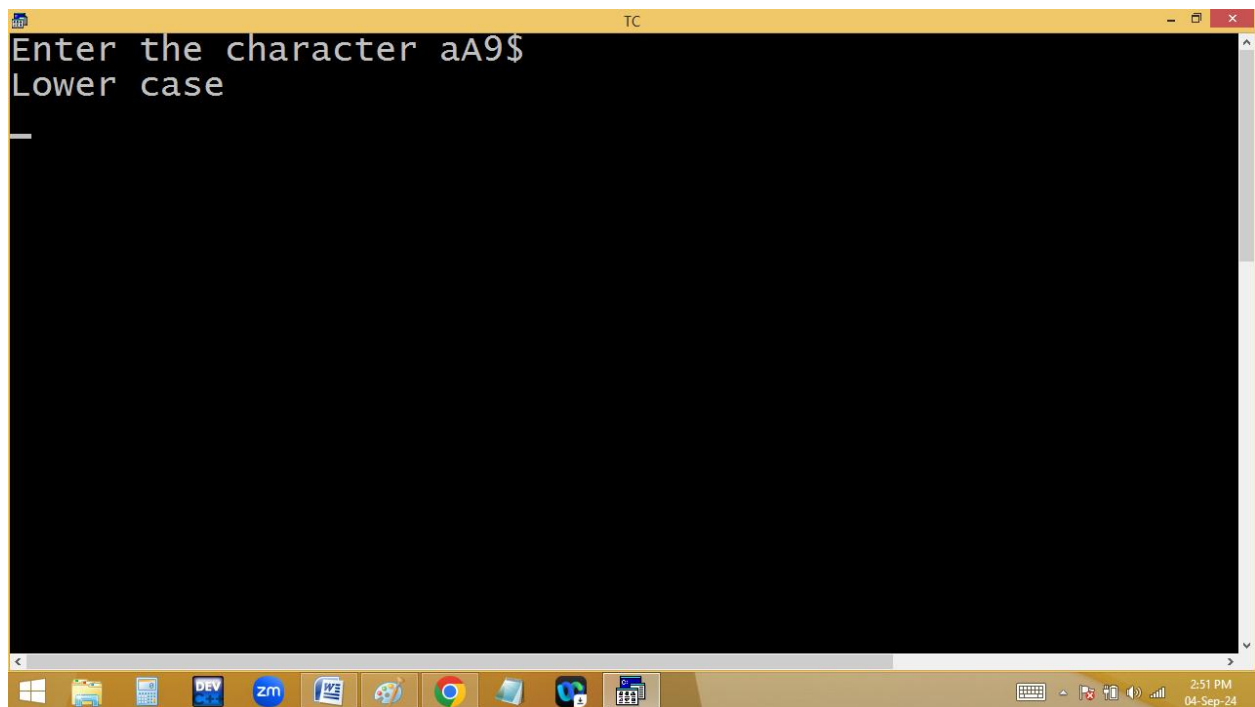


The screenshot shows the Turbo C++ (TC) console window. The prompt 'Enter the character' is displayed, and the user has entered the character 'g'. The program has executed the `if` statements, and the output 'Lower case' is printed on the next line. The console background is black with white text.

```
Enter the character g
Lower case
```





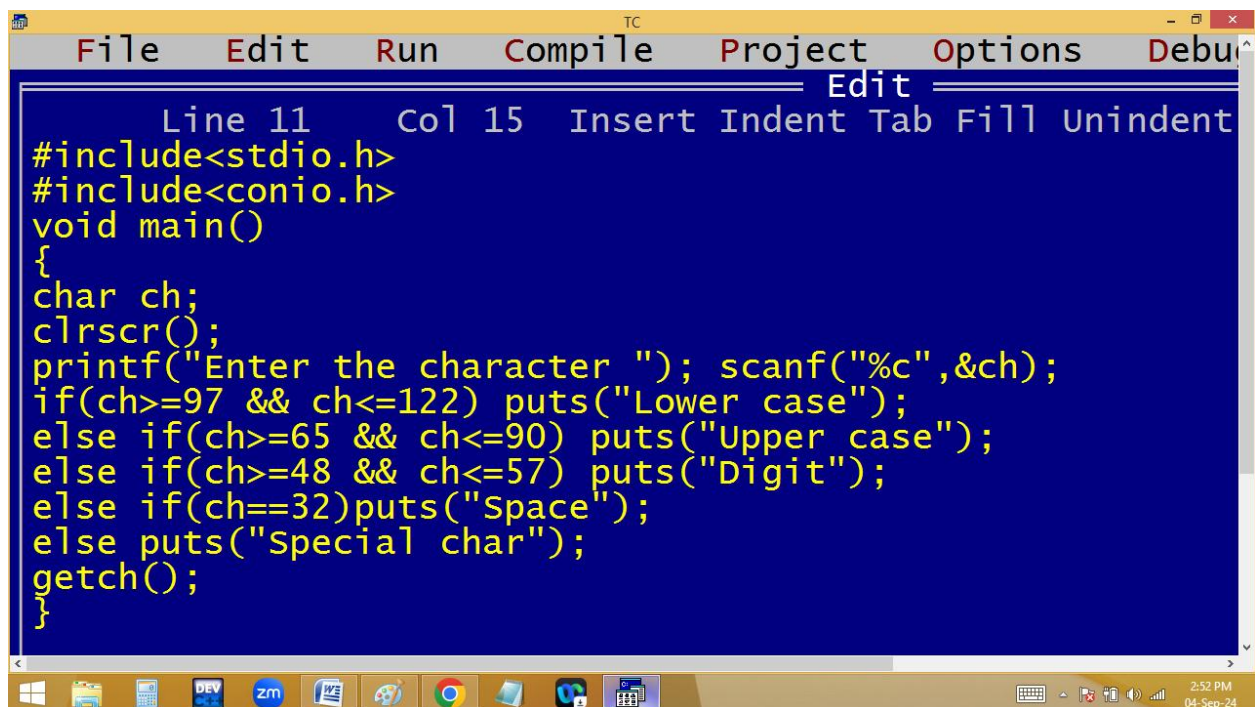


TC

```
Enter the character aA9$
Lower case
```

Windows taskbar at the bottom shows icons for File Explorer, DEV, zm, Word, Chrome, and other applications. The system clock shows 2:51 PM on 04-Sep-24.

Using ascii values:



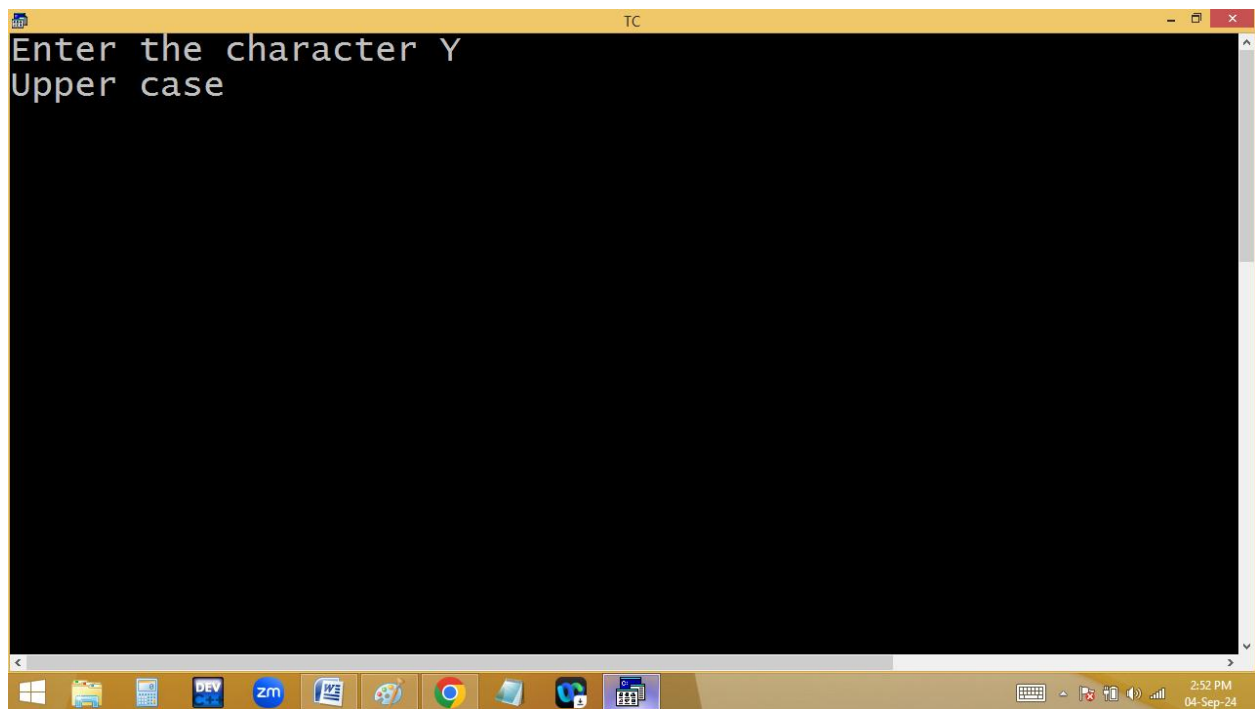
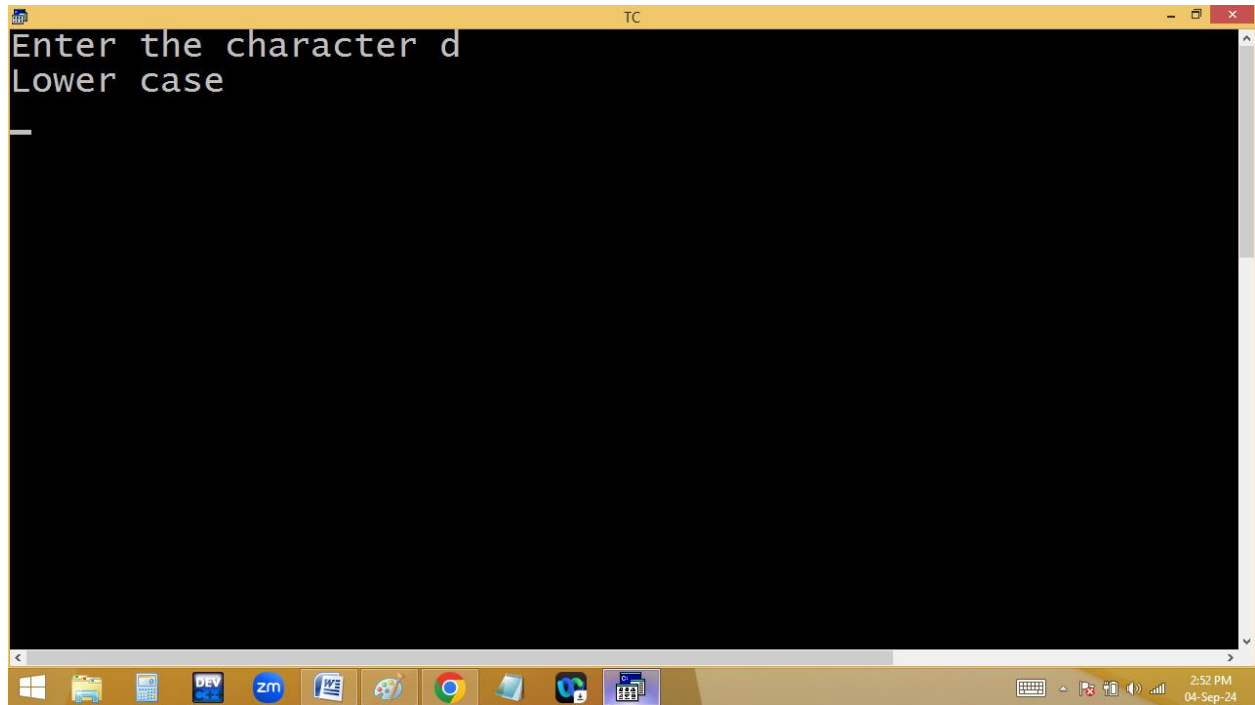
TC

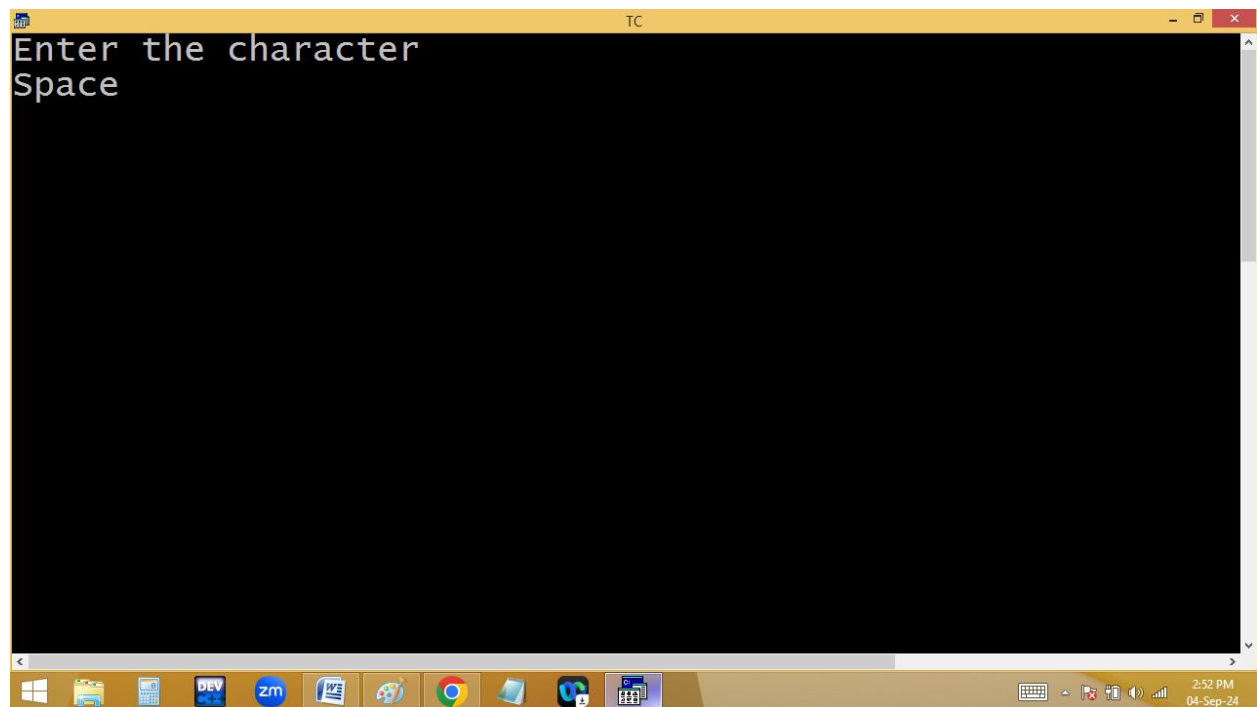
File Edit Run Compile Project Options Debug

Line 11 Col 15 Insert Indent Tab Fill Unindent

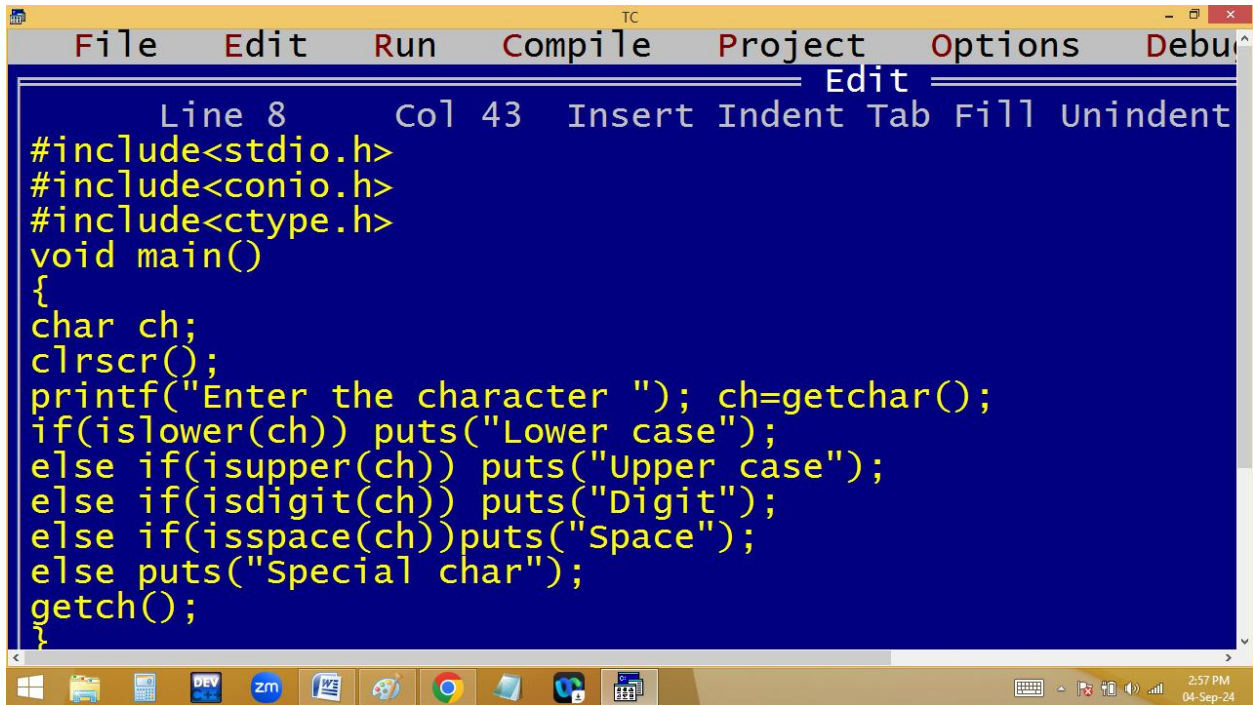
```
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); scanf("%c",&ch);
if(ch>=97 && ch<=122) puts("Lower case");
else if(ch>=65 && ch<=90) puts("Upper case");
else if(ch>=48 && ch<=57) puts("Digit");
else if(ch==32)puts("Space");
else puts("Special char");
getch();
}
```

Windows taskbar at the bottom shows icons for File Explorer, DEV, zm, Word, Chrome, and other applications. The system clock shows 2:52 PM on 04-Sep-24.





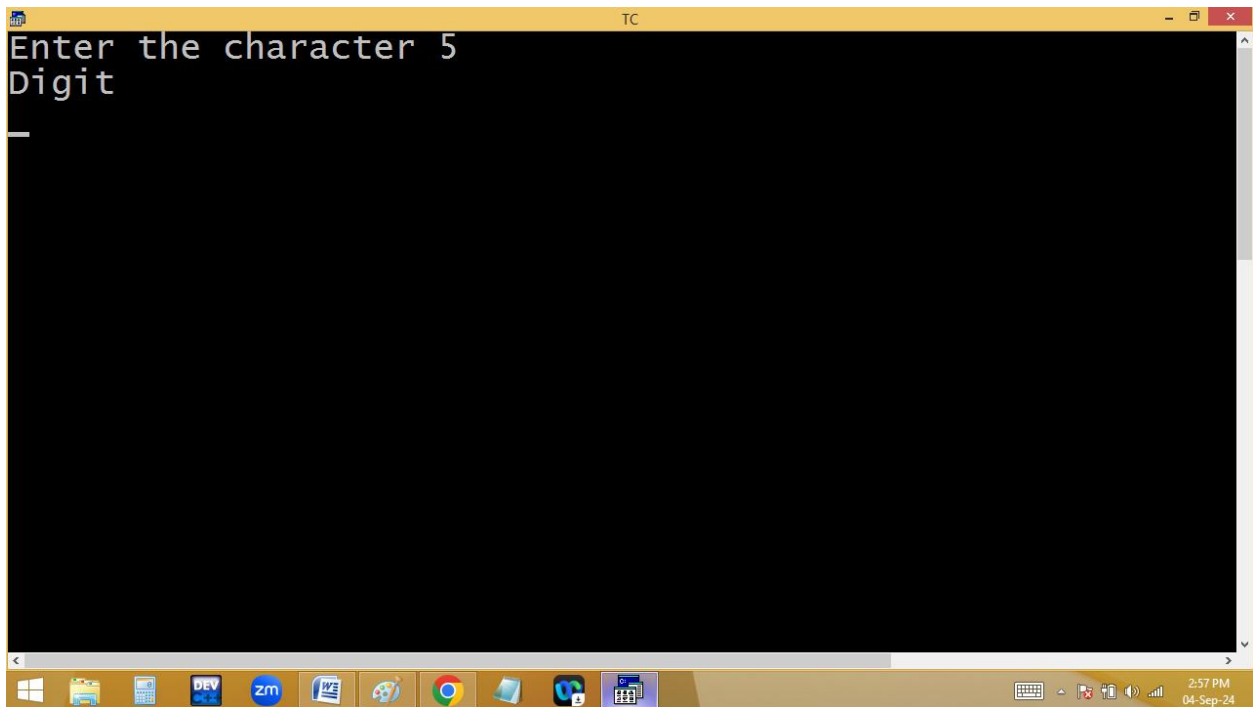
Using predefined functions:



The screenshot shows the Turbo C++ (TC) editor window. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates 'Line 8 Col 43' and provides options like 'Insert', 'Indent', 'Tab', 'Fill', and 'Unindent'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
#include<ctype.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); ch=getchar();
if(islower(ch)) puts("Lower case");
else if(isupper(ch)) puts("Upper case");
else if(isdigit(ch)) puts("Digit");
else if(isspace(ch)) puts("Space");
else puts("Special char");
getch();
}
```

The Windows taskbar at the bottom shows various icons including Windows Explorer, DEV C++, Zoom, and Google Chrome. The system clock in the bottom right corner displays '2:57 PM' and '04-Sep-24'.



The screenshot shows the Turbo C++ console window. The output text is:

```
Enter the character 5
Digit
```

The console window has a black background with white text. The Windows taskbar at the bottom is identical to the first screenshot, showing the same icons and system clock.

getch()	getche()	getchar()
Reads single char	Reads single char	Reads single char
Entered char is not visible	char is visible	char is visible

visible		
Not waiting for enter key	Not waiting for enter key	Waiting for enter key
Belongs to conio.h	Belongs to conio.h	Belongs to stdio.h
char not changeable	char not changeable	char changeable

Note: All the above functions returns ASCII value(int) of reading character.

isupper(), islower(), isdigit(), isspace():

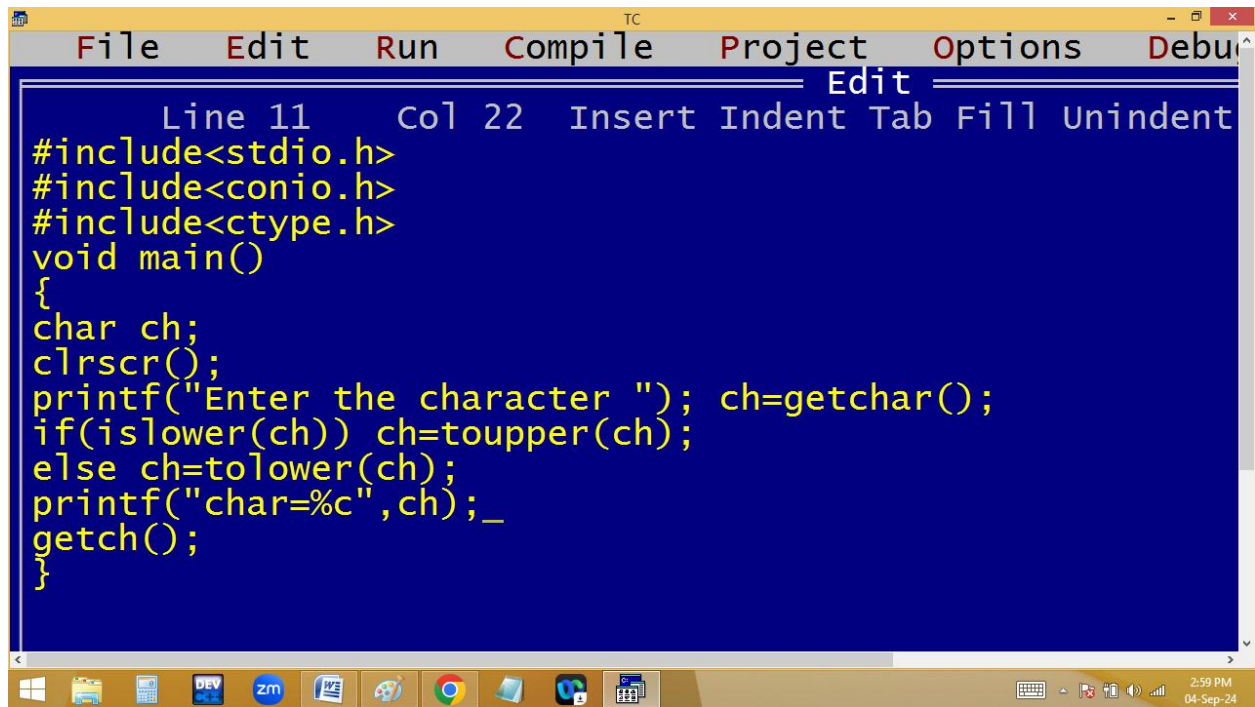
They are used to check the given character is upper / lower / digit / space.

tolower(), toupper(): They are used to convert the character from upper case to lower and lower to upper.

All these functions / **macros** available in **<ctype.h>**

Lower to upper / upper to lower:

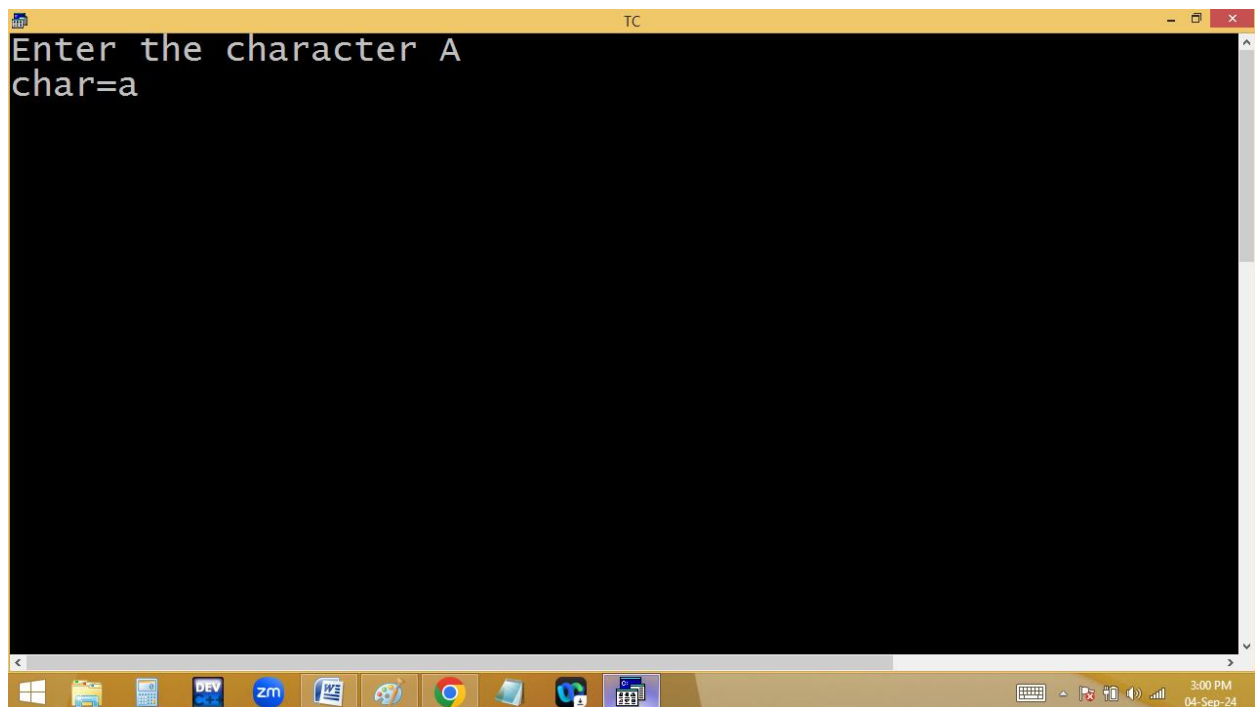
Using predefined functions:



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug) and a toolbar. The main editing area has a blue background and contains the following C code:

```
Line 11   Col 22   Insert  Indent  Tab  Fill  Unindent
#include<stdio.h>
#include<conio.h>
#include<ctype.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); ch=getchar();
if(islower(ch)) ch=toupper(ch);
else ch=tolower(ch);
printf("char=%c",ch);_
getch();
}
```

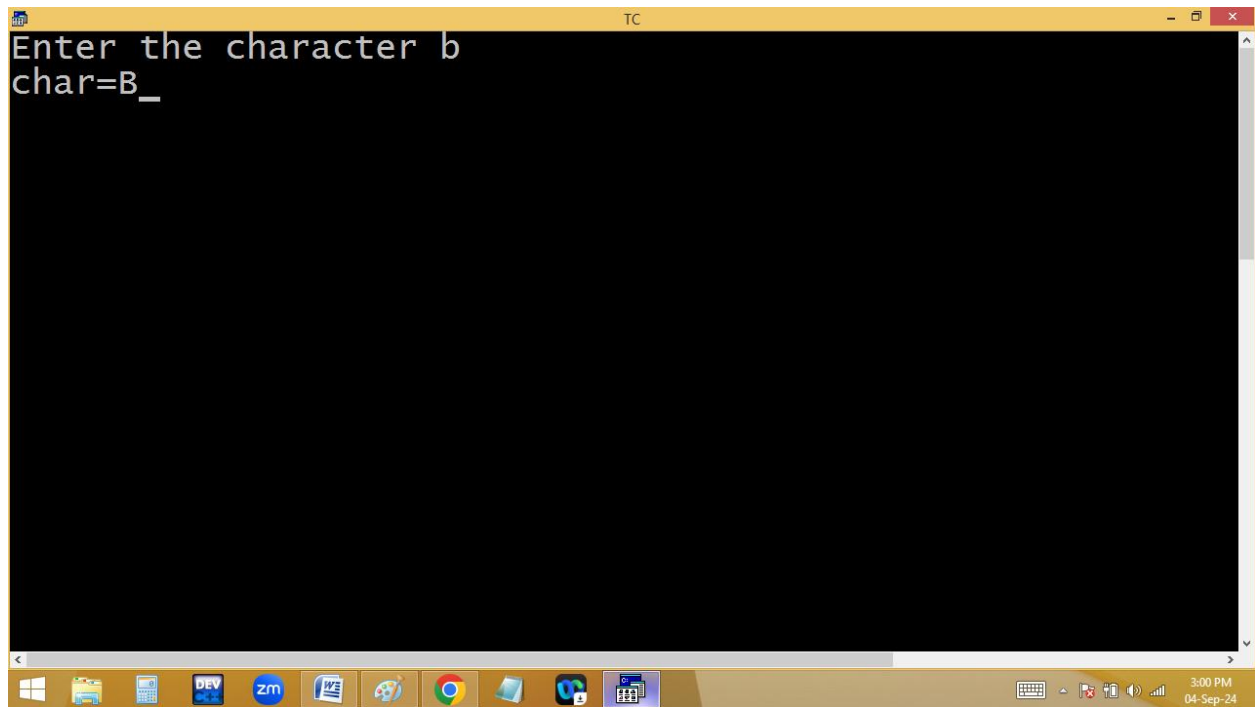
The Windows taskbar at the bottom shows various application icons and the system clock indicating 2:59 PM on 04-Sep-24.



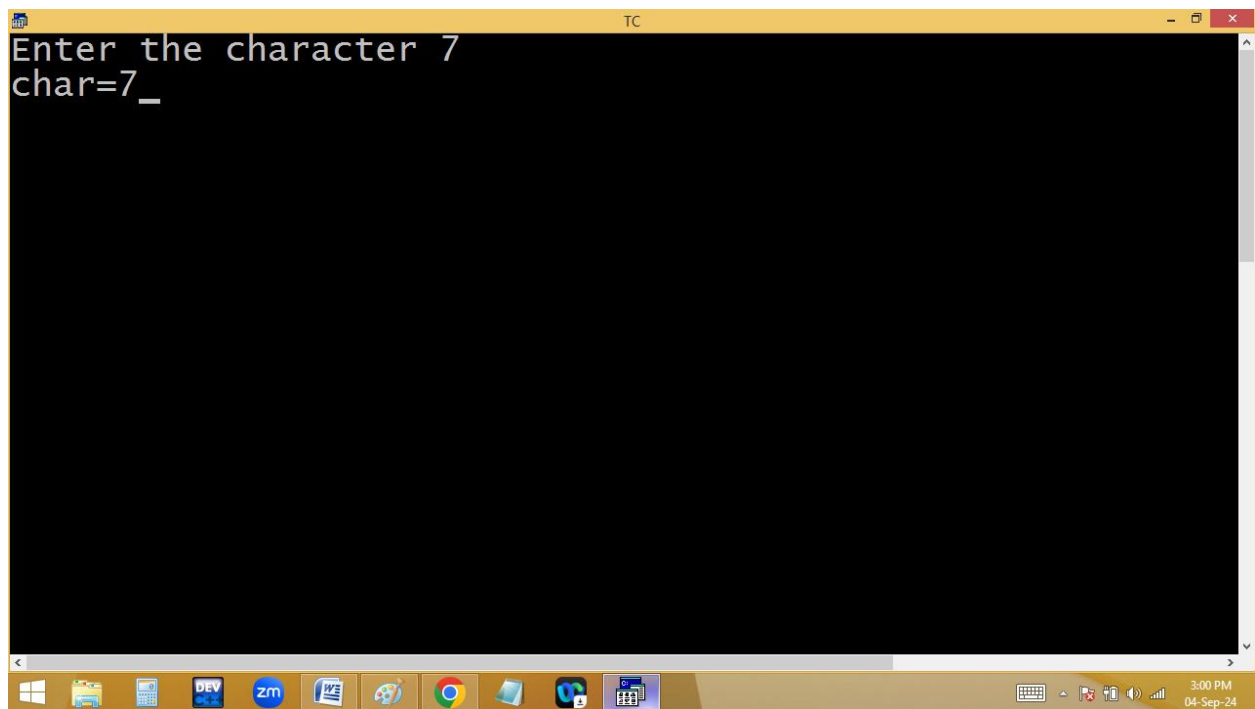
The screenshot shows the Turbo C++ (TC) IDE after the program has been executed. The main editing area now has a black background and displays the program's output:

```
Enter the character A
char=a
```

The Windows taskbar at the bottom shows the same application icons, and the system clock indicates 3:00 PM on 04-Sep-24.



```
TC
Enter the character b
char=B_
```

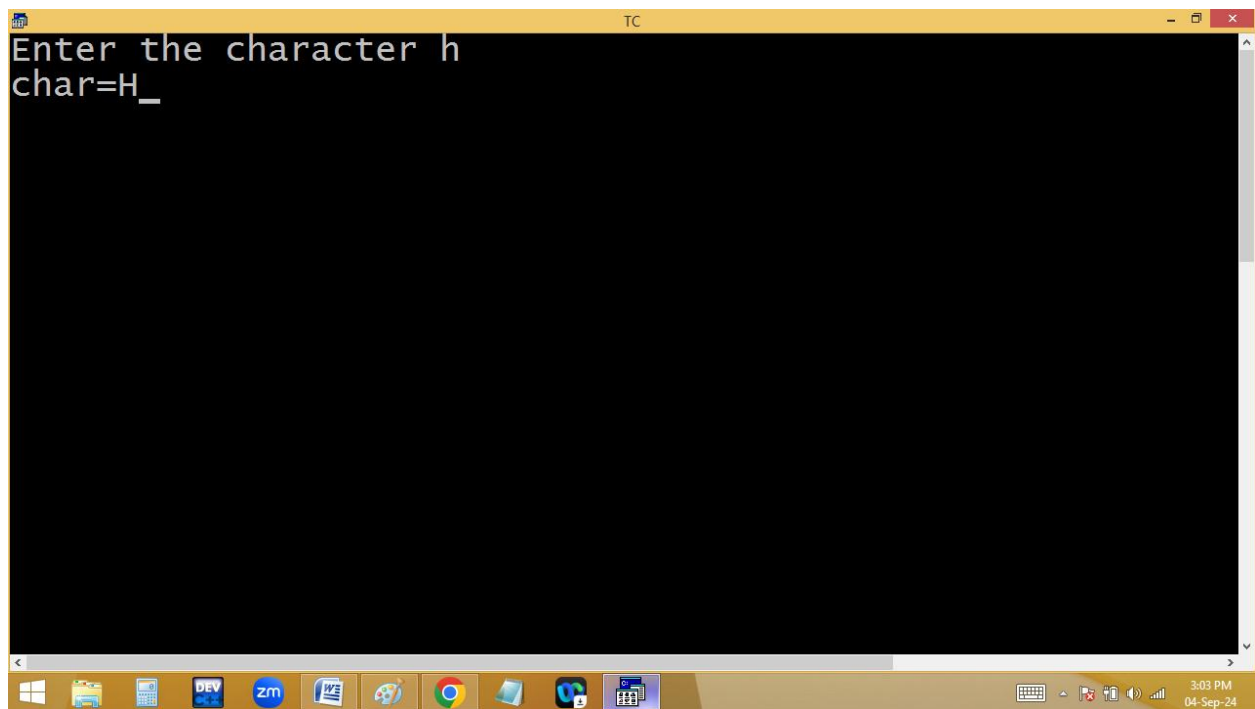
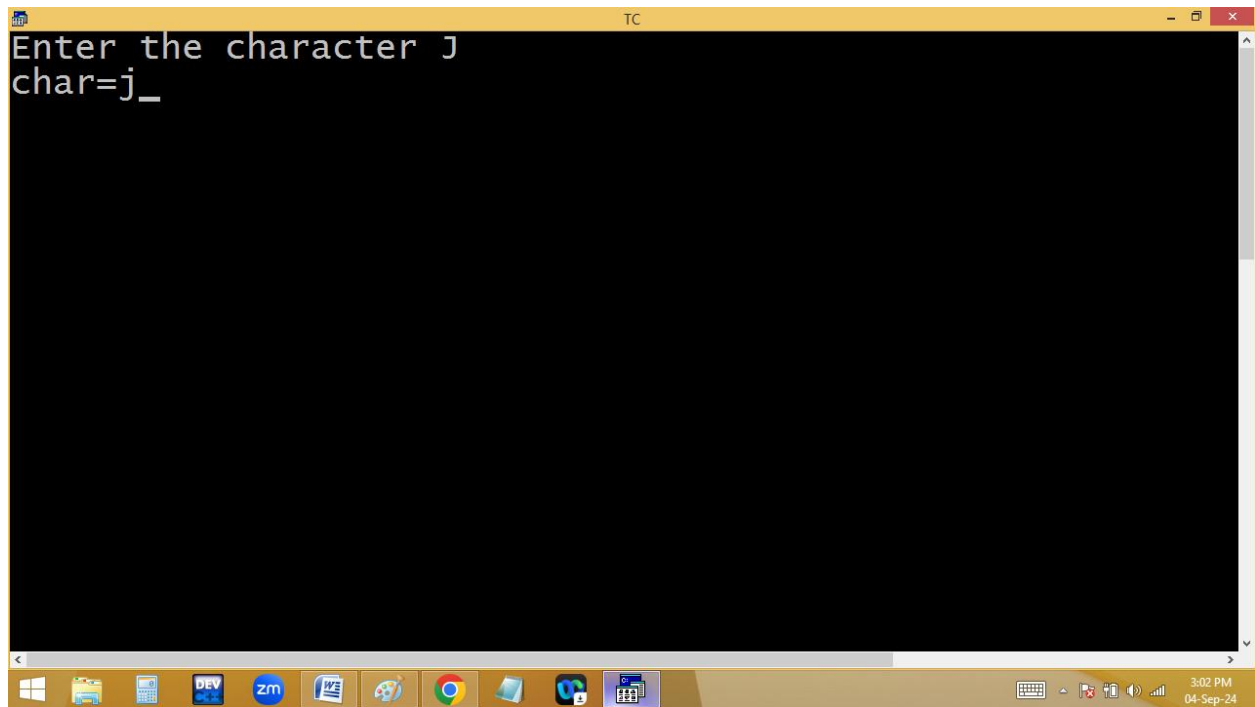


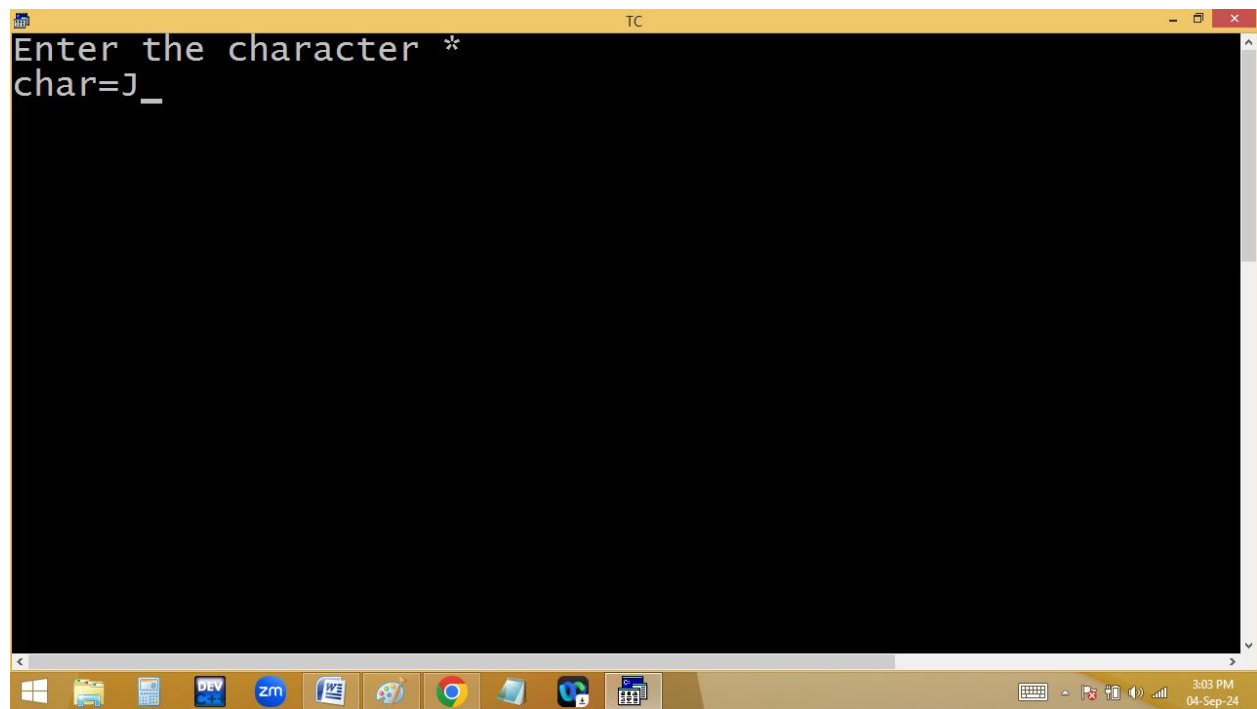
```
TC
Enter the character 7
char=7_
```

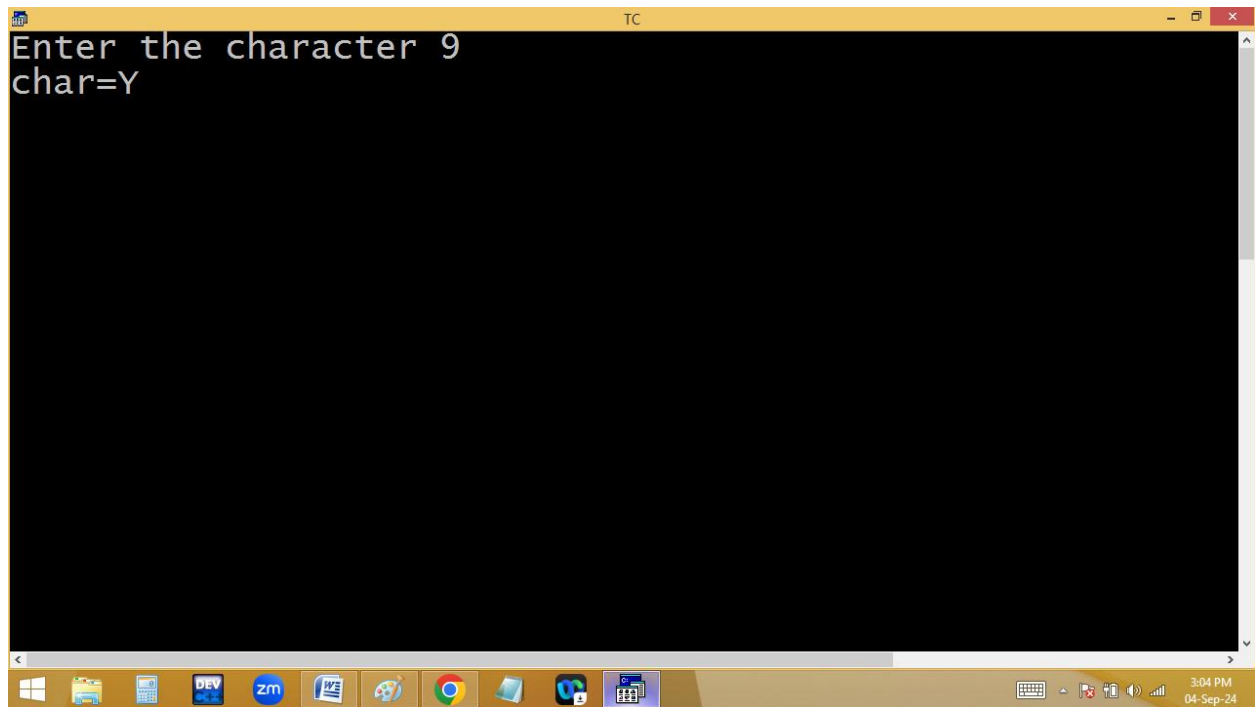
Without using predefined functions:

$$\begin{array}{r} a = 97 \\ - 32 \\ \hline 65 = A \\ + 32 \\ \hline a = 97 \end{array}$$

```
TC
File Edit Run Compile Project Options Debug
Edit
Line 9 Col 29 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); scanf("%c",&ch);
if(ch>='a'&&ch<='z') ch-=32; /* ch=ch-32; */
else ch+=32; /* ch=ch+32; */
printf("char=%c",ch);
getch();
}
```

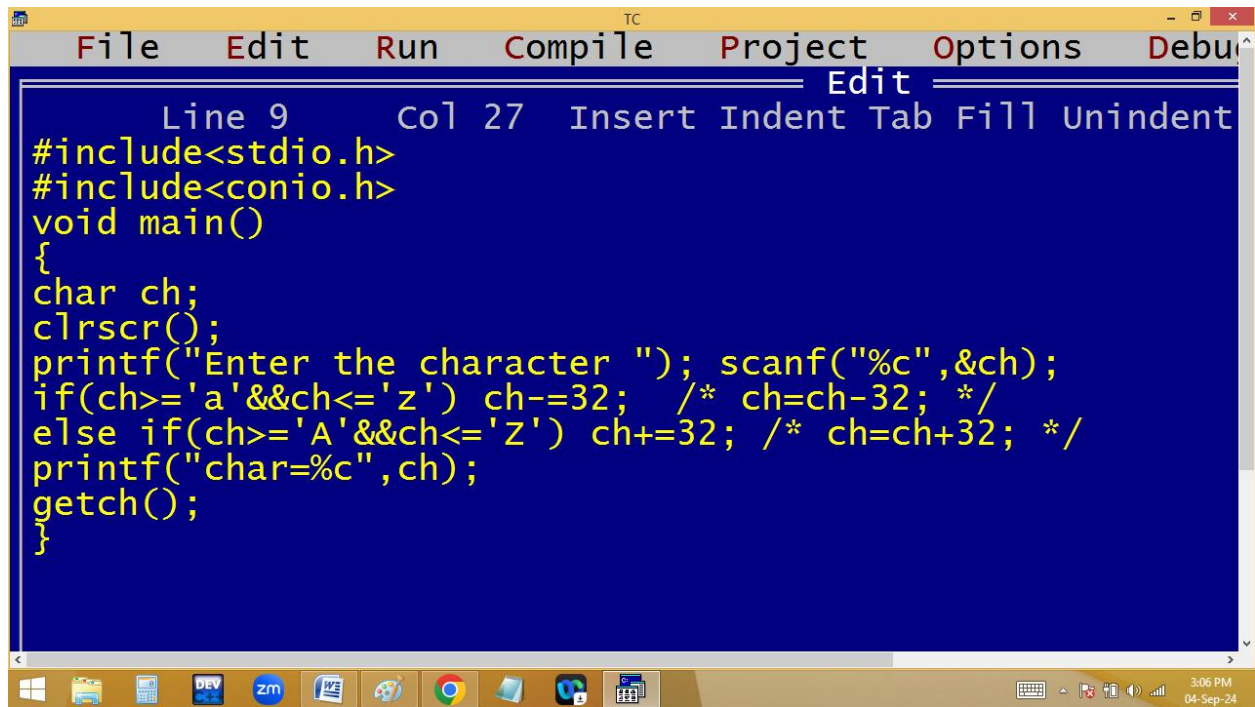






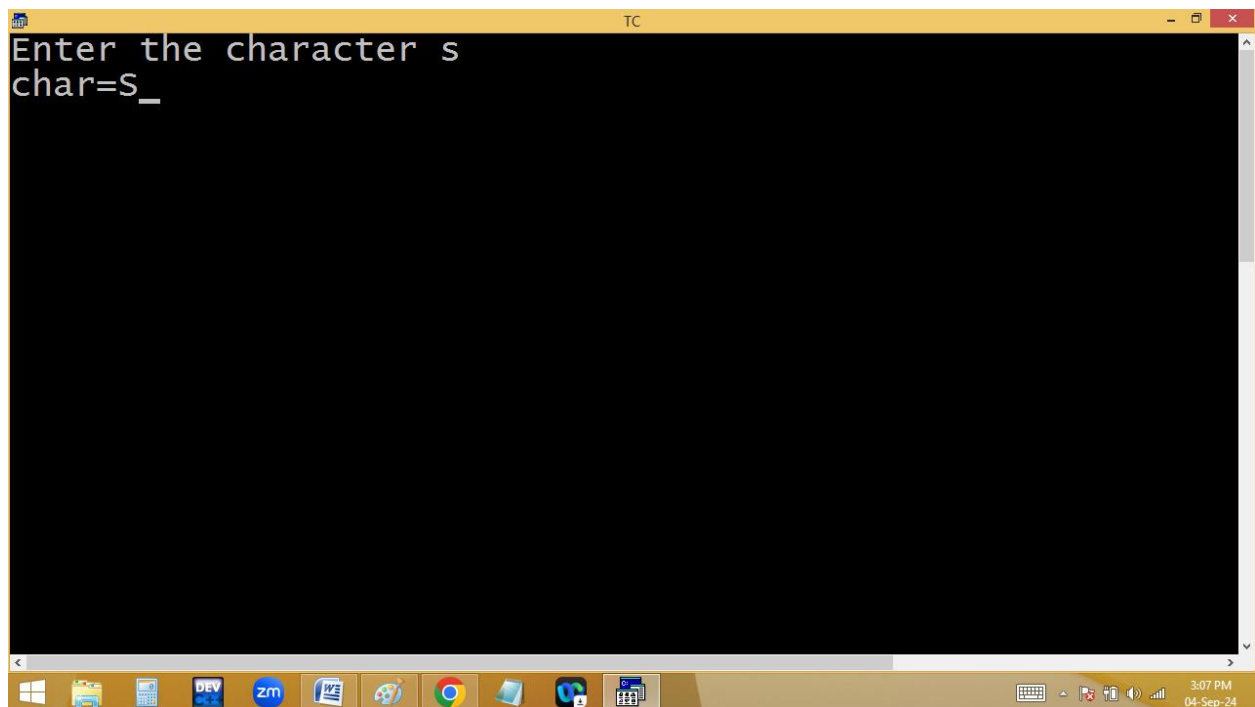
$$\begin{array}{r} * = 42 \\ 32 \\ \hline 74 - J \end{array}$$

$$\begin{array}{r} 9 = 57 \\ + 32 \\ \hline 89 = Y \end{array}$$



The screenshot shows the Turbo C++ (TC) editor window. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates 'Line 9 Col 27' and provides options like Insert, Indent, Tab, Fill, and Unindent. The code is written in yellow text on a blue background. It includes `<stdio.h>` and `<conio.h>`, defines a `main` function, declares a `char ch`, and uses `clrscr()` to clear the screen. It prompts the user to 'Enter the character ' and reads a character into `ch` using `scanf`. A conditional check `if(ch>='a'&&ch<='z')` determines if the character is lowercase; if so, it subtracts 32 (`ch-=32`) to convert it to uppercase. Otherwise, it adds 32 (`ch+=32`) to convert it to lowercase. Finally, it prints the result with `printf("char=%c",ch)` and waits for a key press with `getch()`.

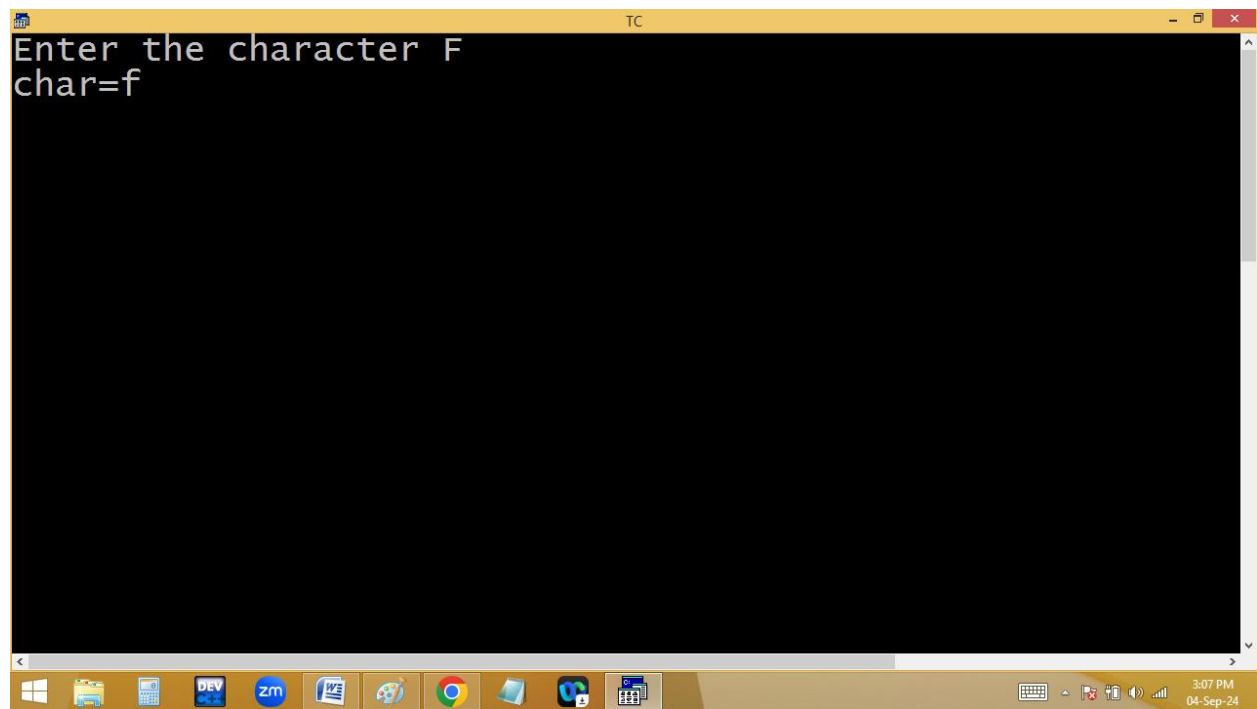
```
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); scanf("%c",&ch);
if(ch>='a'&&ch<='z') ch-=32; /* ch=ch-32; */
else if(ch>='A'&&ch<='Z') ch+=32; /* ch=ch+32; */
printf("char=%c",ch);
getch();
}
```



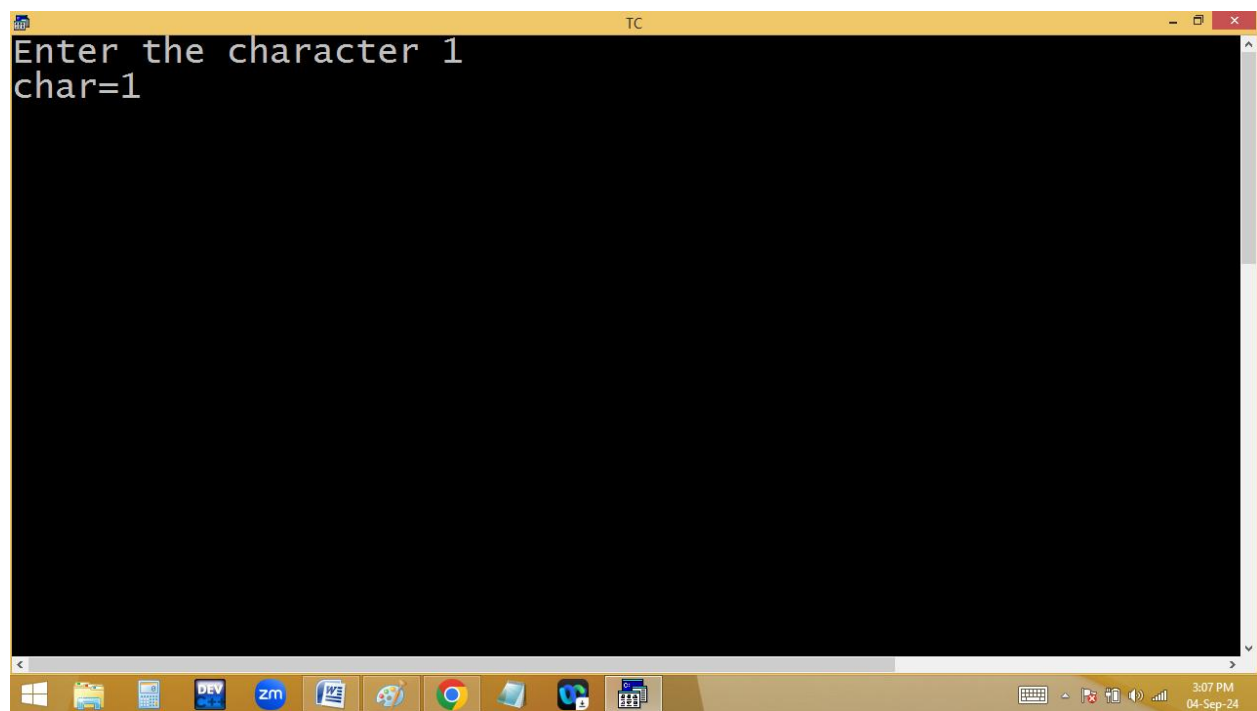
The screenshot shows the Turbo C++ console window. The prompt 'Enter the character ' is displayed, and the user has entered the character 's'. The output shows 'char=S_', indicating that the lowercase 's' has been converted to uppercase 'S'. The status bar at the bottom right shows the time as 3:07 PM on 04-Sep-24.

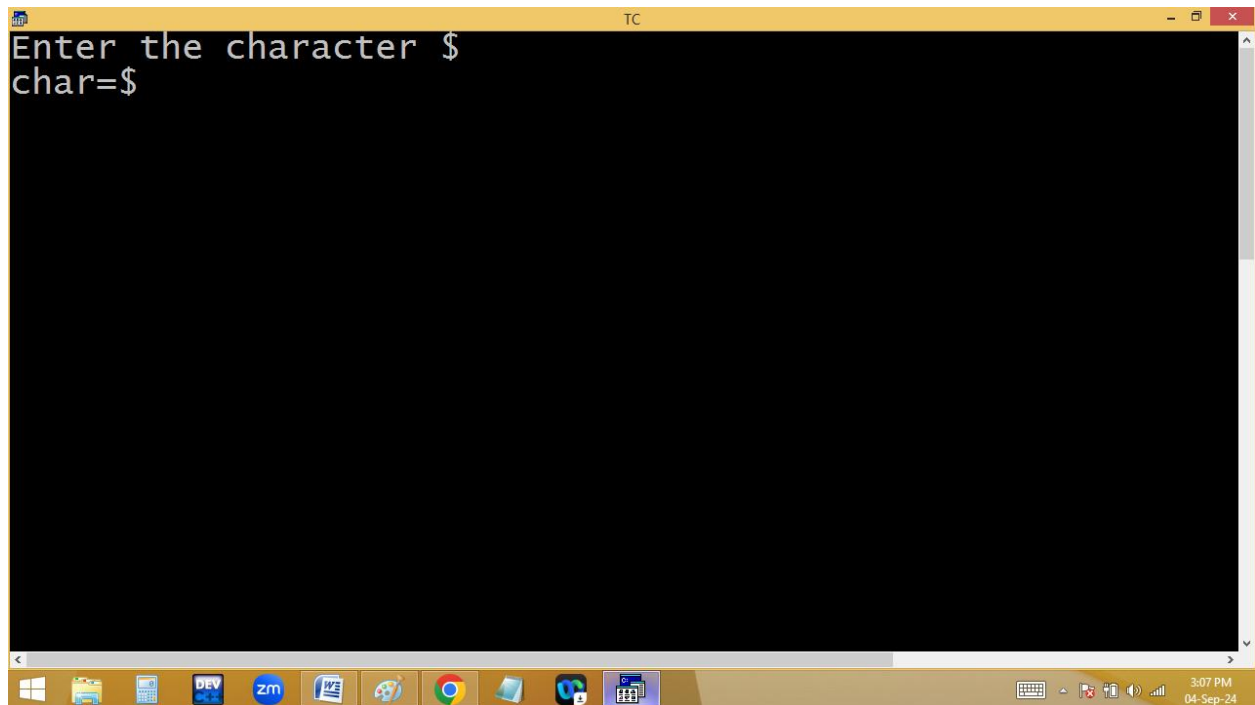
```
Enter the character s
char=S_
```

```
TC
Enter the character F
char=f
```



```
TC
Enter the character 1
char=1
```





```
TC
Enter the character $
char=$
```

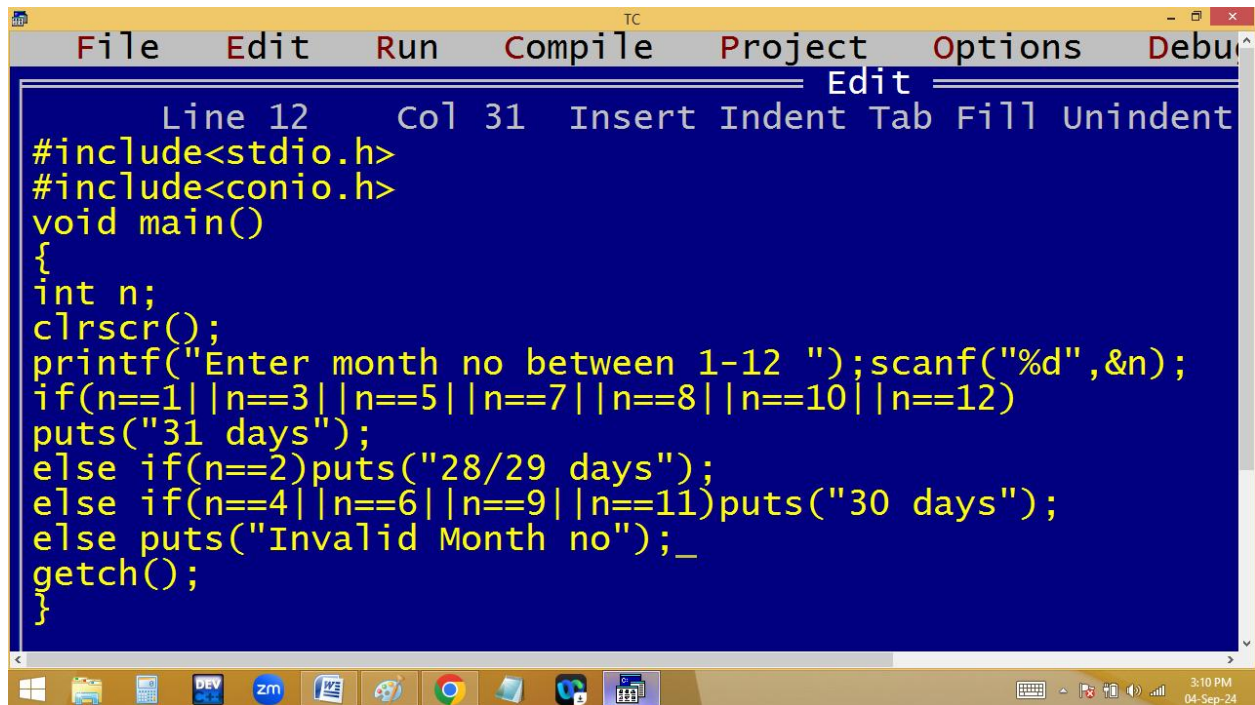
Finding no of days in given month:

1,3,5,7,8,10,12 – 31 days

4,6,9,11 – 30 days

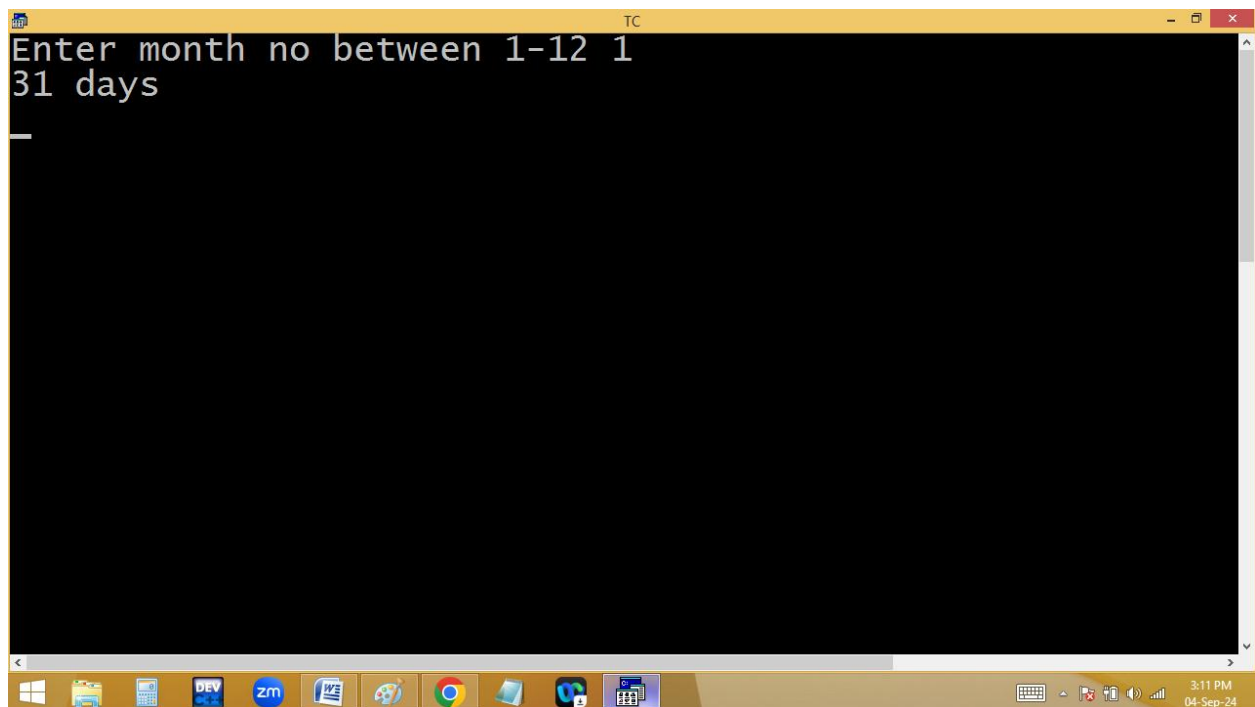
2 – 28/29 days

Others – invalid month



The screenshot shows the Turbo C++ (TC) editor window. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates 'Line 12 Col 31' and provides editing options: Insert, Indent, Tab, Fill, and Unindent. The code is written in yellow text on a blue background. It includes headers for stdio.h and conio.h, and defines a main function. The program prompts the user to enter a month number (1-12) and uses a series of if-else statements to determine the number of days in that month. For months 1, 3, 5, 7, 8, 10, and 12, it prints '31 days'. For month 2, it prints '28/29 days'. For months 4, 6, 9, and 11, it prints '30 days'. For any other input, it prints 'Invalid Month no'. The program ends with a getch() call to pause execution.

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int n;
    clrscr();
    printf("Enter month no between 1-12 ");scanf("%d",&n);
    if(n==1||n==3||n==5||n==7||n==8||n==10||n==12)
        puts("31 days");
    else if(n==2)puts("28/29 days");
    else if(n==4||n==6||n==9||n==11)puts("30 days");
    else puts("Invalid Month no");_
    getch();
}
```

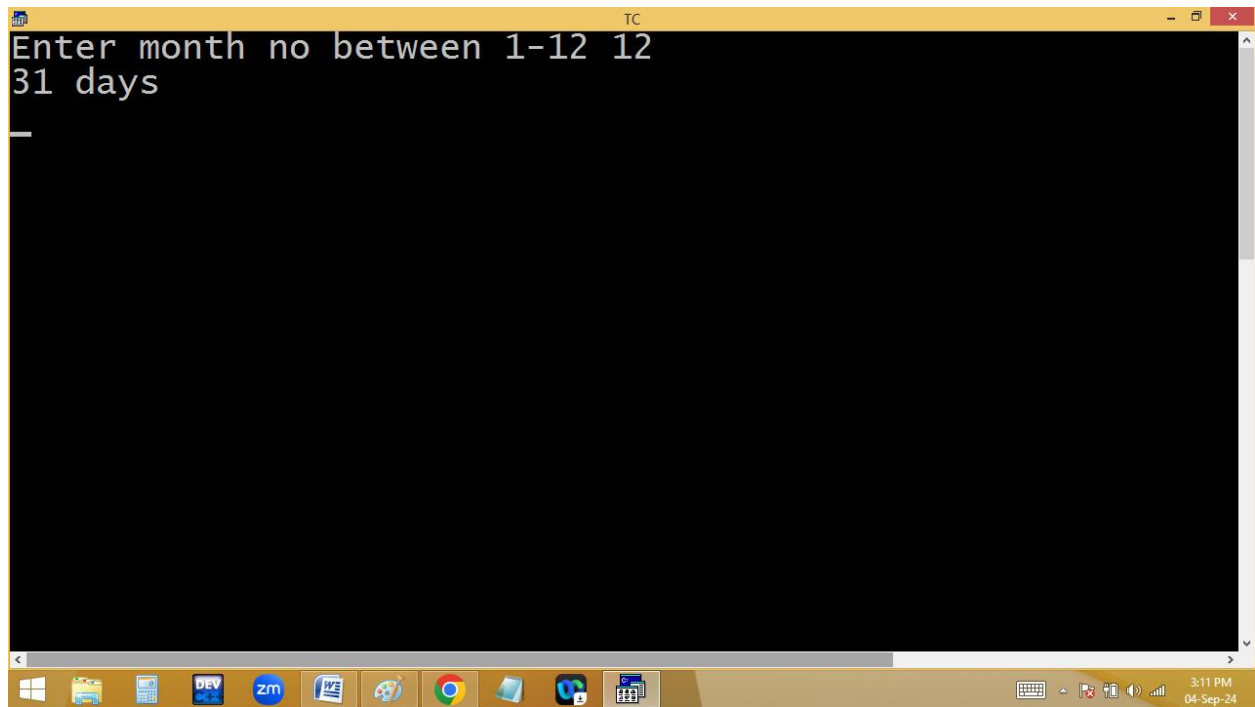


The screenshot shows the Turbo C++ (TC) console window. The program has been executed, and the output is displayed in white text on a black background. The first line shows the prompt 'Enter month no between 1-12' followed by the user input '1'. The second line shows the output '31 days'. The console window has a standard Windows taskbar at the bottom with various application icons and a system clock showing 3:11 PM on 04-Sep-24.

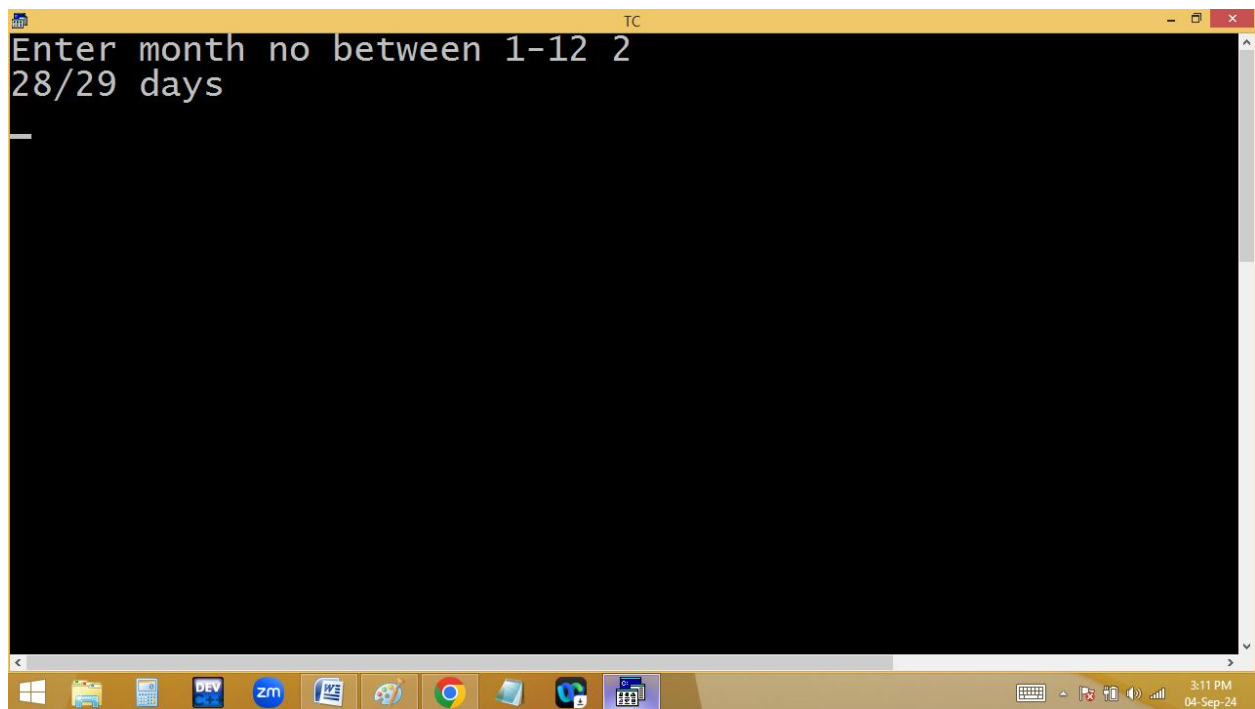
```
Enter month no between 1-12 1
31 days
```

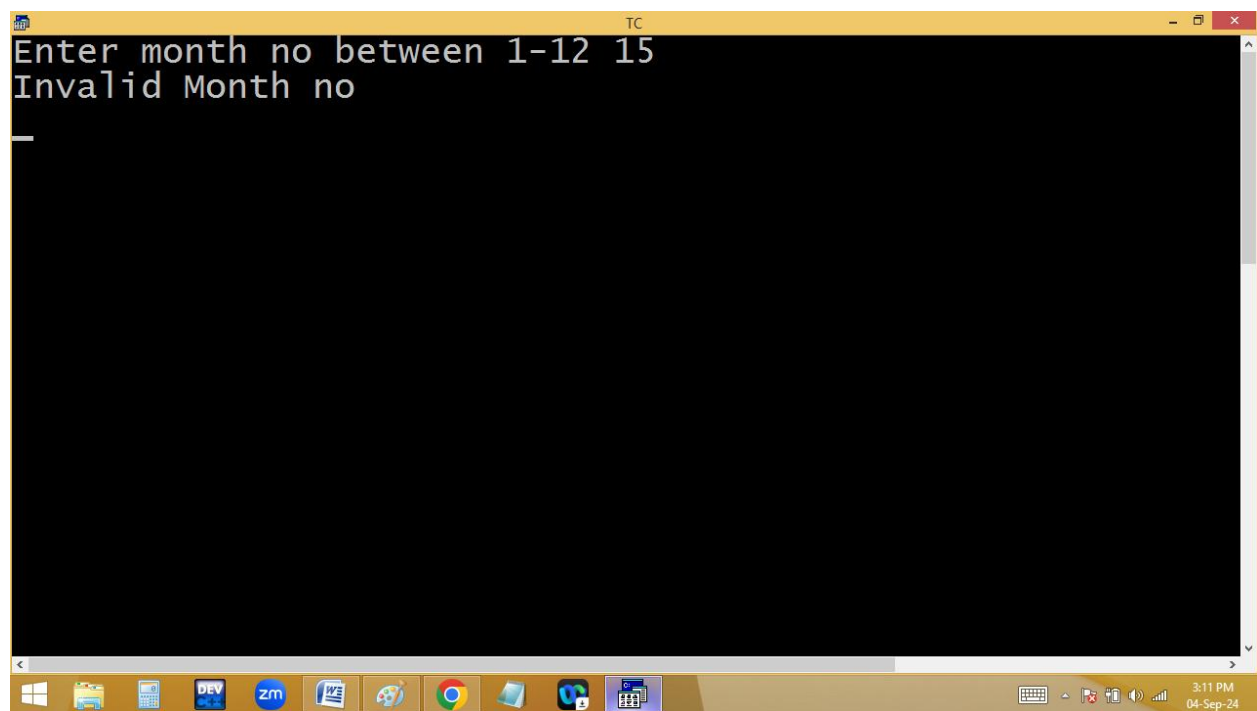
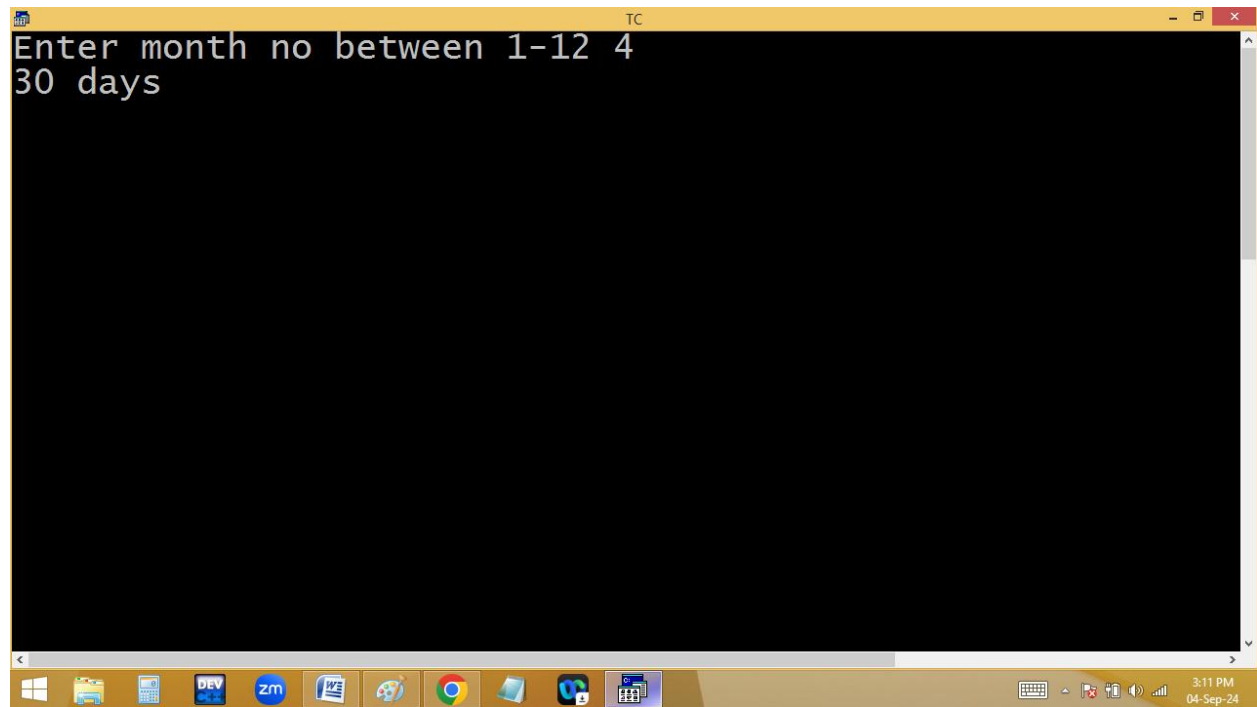


```
TC
Enter month no between 1-12 12
31 days
```

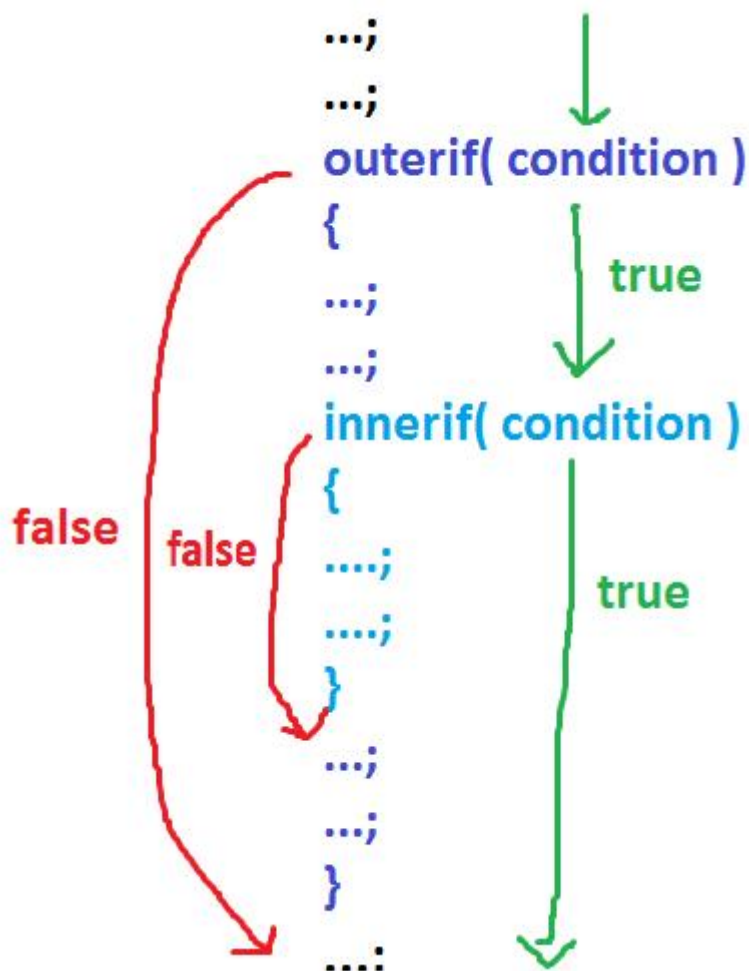


```
TC
Enter month no between 1-12 2
28/29 days
```





Nested if: if within if is called nested if.

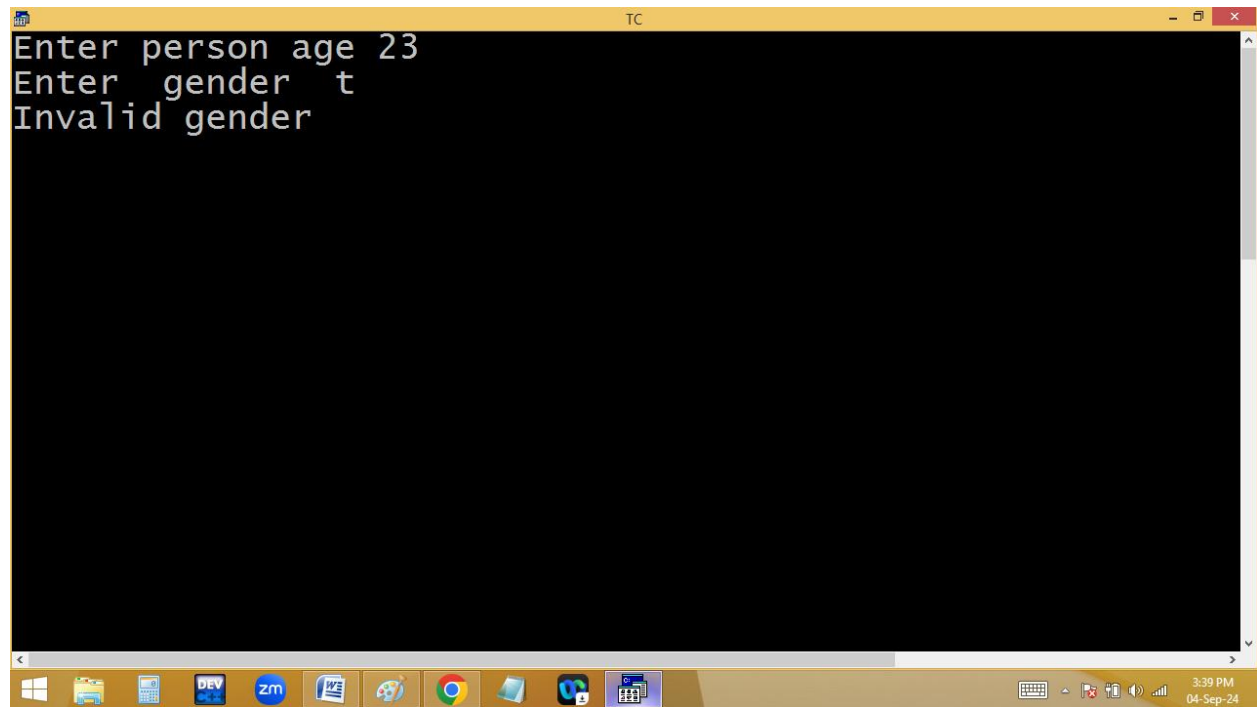


Read a person age and gender and determine that person is eligible for marriage or not.

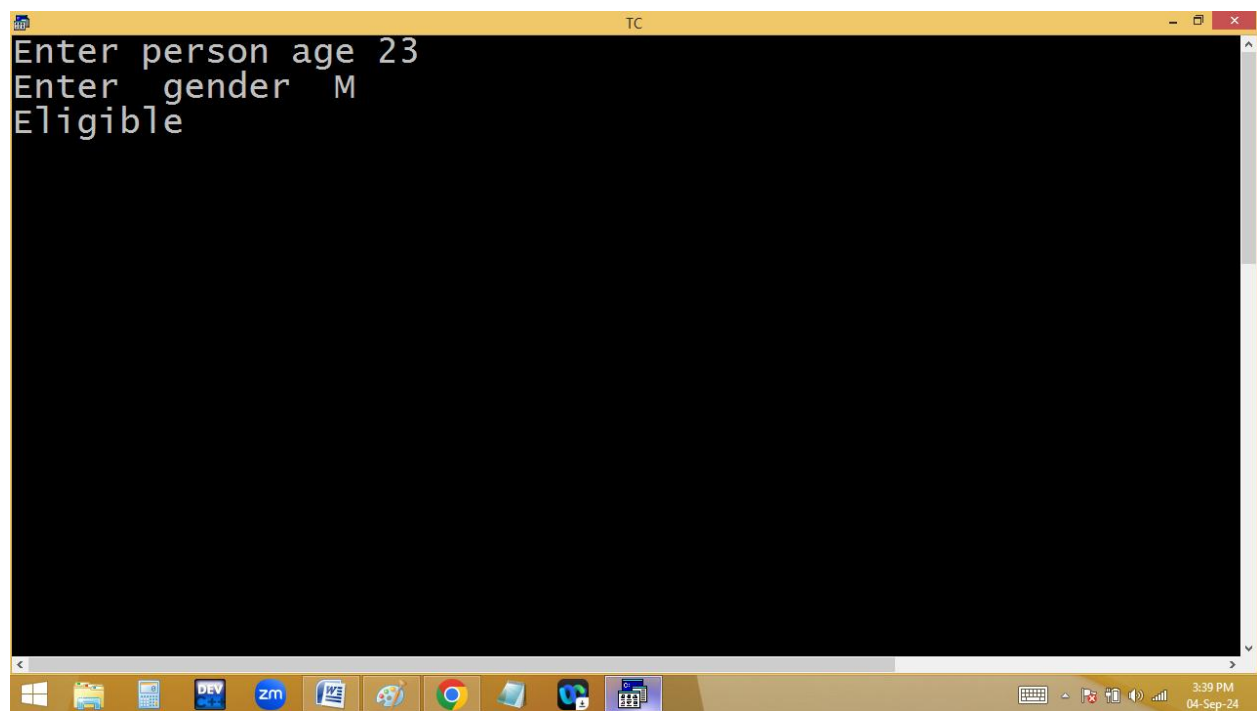
```
TC
#include<stdio.h>
#include<conio.h>
void main()
{
int age; _char gen;
clrscr();
printf("Enter person age ");scanf("%d",&age);
printf("Enter gender "); flushall(); scanf("%c",&gen);
if(gen=='m' || gen=='M' || gen=='f' || gen=='F')
{
if(age>=21 || (gen=='f' || gen=='F')&&age>=18)
puts("Eligible");
else puts("Not Eligible");
}
else puts("Invalid gender");
getch();
}
```

```
TC
Enter person age 22
Enter gender g
Invalid gender
```

```
TC
Enter person age 23
Enter gender t
Invalid gender
```

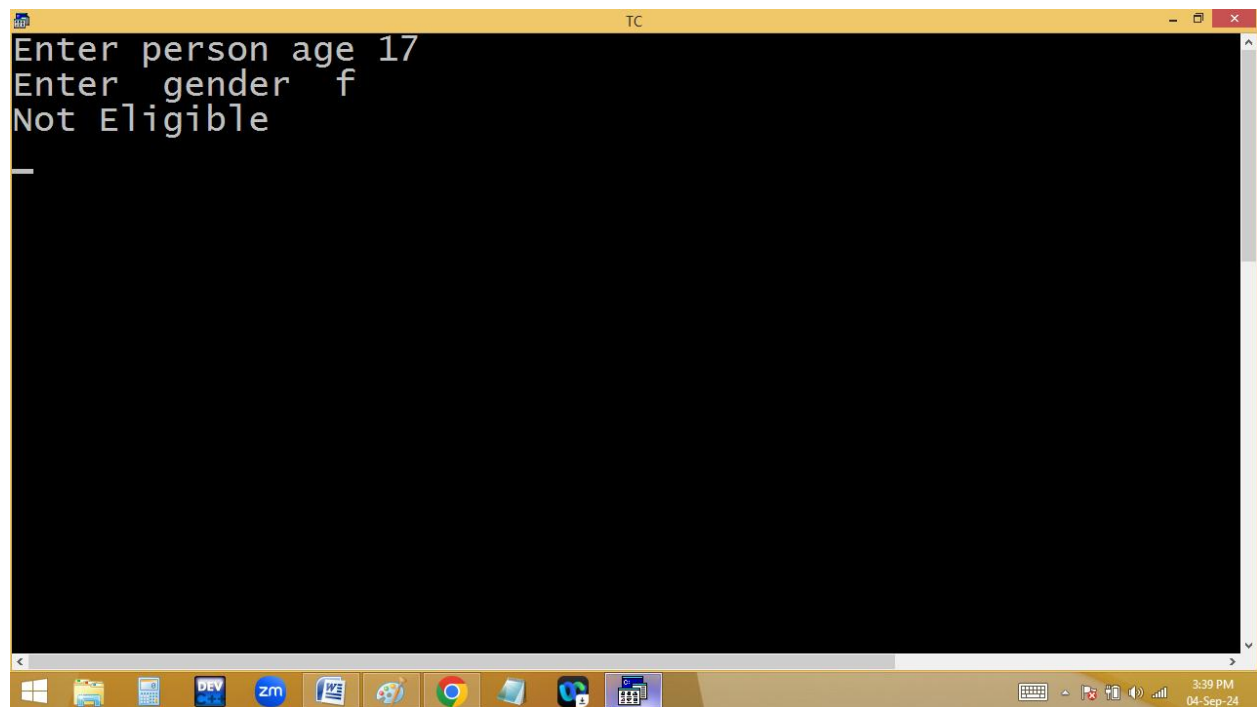


```
TC
Enter person age 23
Enter gender M
Eligible
```



```
TC
Enter person age 20
Enter gender m
Not Eligible
```

```
TC
Enter person age 20
Enter gender f
Eligible
```



```
TC
Enter person age 17
Enter gender f
Not Eligible
```

ATM PIN Validation:

```
TC
#include<stdio.h>
#include<conio.h>
void main()
{
int  pin,count=0;
clrscr();
abc:
printf("Enter pin no "); scanf("%4d",&pin);
if(pin==1234)puts("welcome to HDFC ATM");
else
{
count++;
if(count==3)puts("Ur card blocked for 24 hours");
else {puts("Invalid pin"); goto abc;}
}
getch();
}
```

```
TC
Enter pin no 1234
Welcome to HDFC ATM
```



```
TC
Enter pin no 1111
Invalid pin
Enter pin no 2222
Invalid pin
Enter pin no 4443
Ur card blocked for 24 hours
```

```
TC
Enter pin no 123456
Welcome to HDFC ATM
```

